INFOMINA BERHAD

SOFTWARE DESIGN DOCUMENT (SDD) NEW COMMUNICATIONS EQUIPMENT FRAMEWORK (NCEF)

VERSION 1.4

MALAYSIAN COMMUNICATIONS AND MULTIMEDIA COMMISSION (MCMC)

©Infomina Berhad

DOCUMENT AUTHORISATION

The enclosed document has been prepared, reviewed and approved as follows:

For and on behalf of **Infomina Berhad**

Prepared by:			**Approved by:**		
Name	:	En. Hamzah Hamizan	Name	:	Pn. Marina Ibrahim
Designation	:	Business Analyst	Designation	:	Project Manager
Date	:	02/04/2026	Date	:	

For and on behalf of **Malaysian Communications and Multimedia Commission (MCMC)**

Reviewed By:			**Reviewed by:**		
		Signed by:NOR AIN BINTI RAMLI Signed at:2026-04-01 15:35:11 +08:00 Reason:I approve this document			MUHAMAD TAJUZZAMAN BIN MUHAMAD TAMYEZ
Name	:	Pn. Nor Ain Ramli	Name	:	En. Muhamad Tajuzzaman Muhamad Tamyez
Designation	:	Assistant Director	Designation	:	Head of Department
Date	:	01/04/26	**Date**	:	01/04/26

Reviewed By:			**Approved by:**		
Name	:	Ts. Aiman Hilmi Azhar	Name	:	Tn. Abdul Karim Abdul Razak
Designation	:	Head of Department	Designation	:	Head of Technology and Standards Division
Date	:	30 March 2026	Date	:	31 March 2026

VERSION HISTORY

Version	Date	Author	Description
V1.0	20 February 2026	Hamzah Hamizan	SDD Created
V1.1	24 February 2026	Hamzah Hamizan	Updated with consideration from MCMC's feedback on the URS
V1.2	27 February 2026	Hamzah Hamizan	Amended based on discussion held with MCMC on 26 February 2026
V1.3	06 March 2026	Hamzah Hamizan	Amended based on feedback from MCMC on 05 March 2026
V1.4	10 March 2026	Hamzah Hamizan	Appended with additional modules from MCMC on 9 March 2026

Table of Contents

1. Introduction 4
1.1. Purpose 5
1.2. Scope 5
1.3. Intended Audience 5
1.4. Project Background 5
1.5. System Objectives 5
1.6 References 6
1.7 Acronyms and Glossary 7
2. System Overview 8
3. System Architecture 10
3.1. Technology Stack 11
3.2. Deployment Architecture 12
3.3. Logical Architecture 13
3.4. Integration Architecture 14
4. Data Design 15
4.1. Data Model (Entity-Relationship Diagram) 16
4.2. Data Dictionary 16
5. Component Design 18
5.1. User Management Module 19
5.1.1. Purpose 19
5.1.2. Key Functionalities 19
5.1.3. Interfaces 20
5.1.4. Data Requirements 20
5.2. Equipment Registration Module (SDoC) 20
5.2.1. Purpose 20
5.2.2. Key Functionalities 20
5.2.3. Interfaces 22
5.2.4. Data Requirements 22
5.3. Special Approval Module 22
5.3.1. Purpose 23
5.3.2. Key Functionalities 23
5.3.3. Interfaces 24

5.3.4. Data Requirements 24
5.4. Renewal Module 25
5.4.1. Purpose 26
5.4.2. Key Functionalities 26
5.4.3. Interfaces 27
5.4.4. Data Requirements 27
5.5. IMEI/Serial Number Registration Module 27
5.5.1. Purpose 28
5.5.2. Key Functionalities 28
5.5.3. Interfaces 28
5.5.4. Data Requirements 28
5.6. Modification of Registration Module 28
5.6.1. Purpose 29
5.6.2. Key Functionalities 29
5.6.3. Interfaces 29
5.6.4. Data Requirements 29
5.7. Importation Module 30
5.7.1. Purpose 31
5.7.2. Key Functionalities 31
5.7.3. Interfaces 31
5.7.4. Data Requirements 31
5.8. Post-Market Surveillance (PMS) Module 32
5.8.1. Purpose 33
5.8.2. Key Functionalities 33
5.9. Complaint & Surveillance Module 34
5.9.1. Purpose 35
5.9.2. Key Functionalities 35
5.9.3. Interfaces 35
5.9.4. Data Requirements 35
5.10. Compliance Status Management Module 35
5.10.1. Purpose 35
5.10.2. Key Functionalities 35
5.10.3. Interfaces 35
5.10.4. Data Requirements 35
5.10. Public Module 37

5.10.1. Purpose 38
5.10.2. Key Functionalities 38
5.10.3. Interfaces 38
5.10.4. Data Requirements 39
5.11. Mobile Application 39
5.11.1. Purpose 40
5.11.2. Key Functionalities 40
5.11.3. Interfaces 40
5.11.4. Data Requirements 40
5.12. Payment Integration Module 40
5.12.1. Purpose 41
5.12.2. Key Functionalities 41
5.12.3. Interfaces 41
5.12.4. Data Requirements 42
5.13. AI-Enabled Features 42
5.13.1. Purpose 43
5.13.2. Key Functionalities 43
5.13.3. Interfaces 44
5.13.4. Data Requirements 44
5.14. Dashboard & Reporting Module 44
5.14.1. Purpose 45
5.14.2. Key Functionalities 45
5.14.3. Interfaces 45
5.14.4. Data Requirements 46
5.15. Notification Module 46
5.15.1. Purpose 47
5.15.2. Key Functionalities 47
5.15.3. Interfaces 47
5.15.4. Data Requirements 47
5.16. Integration Requirements 47
5.16.1. Purpose 48
5.16.2. Key Functionalities 48
5.16.3. Interfaces 48
5.16.4. Data Requirements 49
5.17. Configuration & Settings 49

5.17.1. Purpose 50
5.17.2. Key Functionalities 50
5.17.3. Interfaces 50
5.17.4. Data Requirements 50
5.18. Data Migration Strategy 51
5.18.1. Purpose 52
5.18.2. Key Functionalities & Phased Approach 52
5.18.3. Interfaces 52
5.18.4. Data Requirements 53
6. HUMAN INTERFACE DESIGN 53
6.1. UI/UX Principles 54
6.2. Key Screen Flows 54
6.3. Wireframes and Mockups 55
7. NON-FUNCTIONAL DESIGN 55
7.1. Performance 55
7.2. Security 55
7.3. Compliance (PDPA) 55
7.4. Availability & Reliability 55
7.5. Backup & Recovery 55
8. REQUIREMENTS TRACEABILITY MATRIX 55
9. APPENDICES 55
Appendix A: User Access Matrix 55
Appendix B: Fee Structure 55
Appendix C: Proposed AI Scoring Mechanism: 55

1. Introduction

1.1. Purpose

This Software Design Document (SDD) provides a comprehensive technical design for the New Communications Equipment Framework (NCEF) online registration system. It translates the business and functional requirements outlined in the User Requirement Specification (URS) [1] into a detailed blueprint for development. This document describes the system architecture, data design, component-level design, and integration strategies that will guide the implementation of the NCEF platform.

1.2. Scope

The scope of this design encompasses all modules and functionalities defined as “In-Scope” in the NCEF URS (v1.2), Section 4. This includes the end-to-end lifecycle of equipment registration, from user onboarding to post-market surveillance activities. The design covers the internal portal for MCMC officers, the public-facing portal for applicants and the general public, native mobile applications, and all required back-end integrations with internal and external systems.

1.3. Intended Audience

This document is intended for the following stakeholders:

- **Development Team:** To understand the technical specifications, architecture, and component design required for implementation.
- **Project Managers:** To oversee the project timeline, resource allocation, and ensure deliverables align with the specified design.
- **QA & Testing Team:** To create test plans and test cases that validate the system against the documented design.
- **System Administrators & DevOps:** To understand the deployment architecture, infrastructure requirements, and configuration for environment setup and maintenance.
- **MCMC Stakeholders:** To review and approve the technical design, ensuring it aligns with their operational and regulatory objectives.

1.4. Project Background

The NCEF is a new regulatory framework established by the Malaysian Communications and Multimedia Commission (MCMC) to modernize and regulate the registration of communications equipment in Malaysia. The development of a centralized online system is the cornerstone of this initiative, replacing legacy systems and manual workflows. This project is aligned with MCMC’s and the nation’s digital transformation goals to enhance public service delivery through efficient, secure, and intelligent technology.

1.5. System Objectives

The primary objectives of the NCEF system design are to:

- **Streamline Operations:** Design an efficient, automated, and user-centric platform that simplifies the end-to-end registration process for all stakeholders.
- **Enhance Regulatory Oversight:** Architect a system with robust data management, audit trails, and reporting capabilities to provide MCMC with comprehensive oversight of the equipment lifecycle.
- **Improve Service Delivery:** Create a transparent, accessible, and responsive system that improves the user experience for applicants and the public.
- **Integrate Seamlessly:** Design a well-defined integration architecture to connect with critical internal (MCMC SIFS) and external (SIRIM, RMCD, SSM, MCMC Pay) systems.
- **Leverage AI:** Incorporate Artificial Intelligence to enhance efficiency and accuracy in document validation, risk assessment, and decision support.
- **Ensure Security and Compliance:** Build a secure, scalable, and reliable platform that complies with the Malaysian Personal Data Protection Act (PDPA) 2010 and best-in-class security standards.

1.6 References

ID	Document	Version	Date
1	NCEF User Requirement Specification (URS)	1.2	Feb 27, 2026
2	NCEF Revised VM Server Request	-	Feb 16, 2026
3	Software Development Design Based on IEEE 1016	-	-

1.7 Acronyms and Glossary

Abbreviation	Description
AAD	Azure Active Directory
AI	Artificial Intelligence
API	Application Programming Interface
DR	Disaster Recovery
ESB	Enterprise Service Bus
FAT	Final Acceptance Test
ICR	Intelligent Character Recognition
IMEI	International Mobile Equipment Identity
KPI	Key Performance Indicator
LLM	Large Language Model
MCM	Management Committee Meeting
MCMC	Malaysian Communications and Multimedia Commission
NCEF	New Communication Equipment Framework
NFR	Non-Functional Requirement
OIC	Officer in Charge
PDPA	Personal Data Protection Act 2010
RCN	Registration Certification Number
RMCD	Royal Malaysian Customs Department
RPO	Recovery Point Objective
RTO	Recovery Time Objective

SDoC	Supplier's Declaration of Conformity
SDD	Software Design Document
SIFS	MCMC's Integrated Financial System
SLA	Service Level Agreement
SSO	Single Sign-On
SSM	Suruhanjaya Syarikat Malaysia (Companies Commission of Malaysia)
UAT	User Acceptance Test
URS	User Requirement Specification
WAF	Web Application Firewall

2. System Overview

The NCEF Online Registration System is a comprehensive, web-based platform designed to manage the entire lifecycle of communications equipment registration in Malaysia. It serves as the single digital interface between MCMC, equipment suppliers (companies, individuals, institutions), consultants, and the general public.

The system is composed of several key portals and applications:

- **Internal Portal:** A secure portal for MCMC officers (System Administrators, OICs, Recommenders, Verifiers, Approvers) to manage applications, configure system settings, monitor operations via dashboards, and perform all back-office functions.
- **External Portal:** A self-service portal for applicants (Suppliers, Principals, Consultants) to register accounts, submit and manage equipment registration applications (SDoC), special approval, renewals, and modifications. It includes a personalized dashboard for tracking application status and managing payments.
- **Public Portal:** An open-access portal for the public to search and verify the certification status of equipment, access regulatory documents (guidelines, standards), view FAQs, and interact with an AI-powered chatbot.
- **Mobile Applications:** Native mobile apps (Android, iOS, Huawei) providing key functionalities on the go, including public search and push notifications for MCMC announcements.

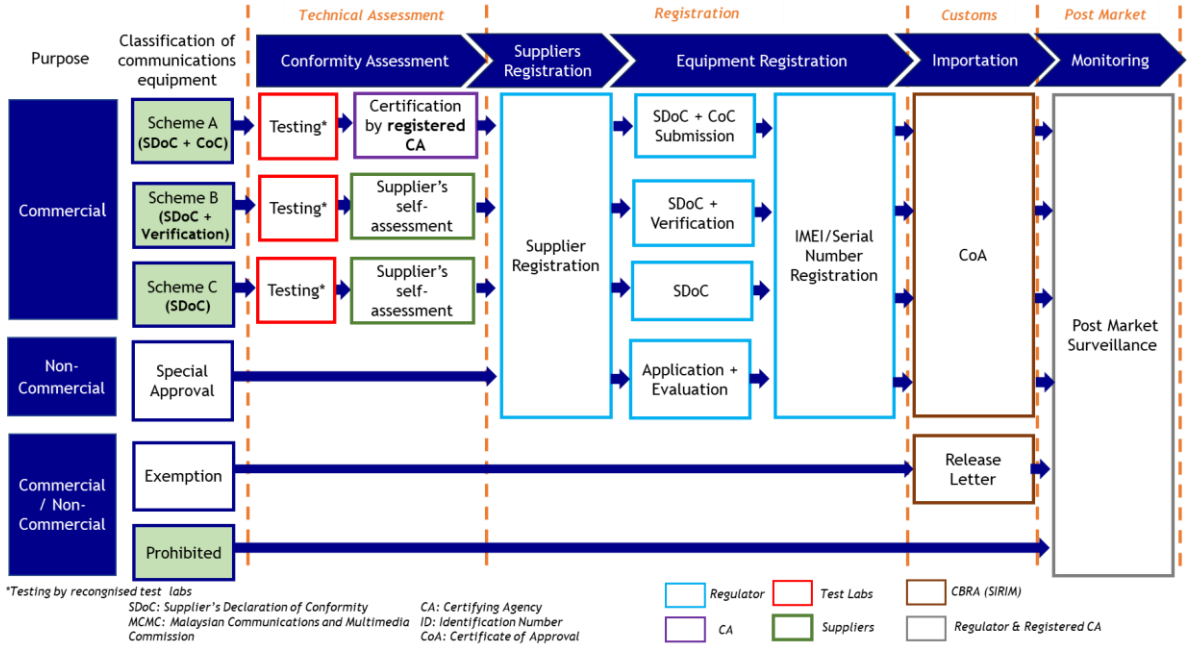


Figure: Overall System Design

The system's core functionality revolves around a set of structured workflows for different registration schemes (Scheme A, B, C) and special approval use cases, with varying levels of scrutiny. A key innovation is the integration of an AI engine to automate document

validation, perform risk scoring, and enable auto-acceptance for low-risk applications, thereby increasing efficiency and allowing MCMC officers to focus on high-value tasks.

Architecturally, the system is designed as a modern, scalable, and secure microservices-based application, leveraging containerization and orchestration technologies to ensure high availability and performance.

3. System Architecture

This section details the overall architecture of the NCEF system, covering the technology stack, deployment model, logical structure, and integration strategy.

3.1. Technology Stack

The NCEF system will be built using a modern, robust, and scalable technology stack, as detailed below:

Layer	Technology	Purpose
Presentation Layer (Web)	ReactJS	A JavaScript library for building interactive, responsive, and bilingual user interfaces for the Web Portal.
Presentation Layer (Mobile)	Flutter	Flutter is an open source framework for building beautiful, natively compiled, multi-platform applications such as Android and iOS from a single codebase, ensuring the build is accessible on all devices.
Application Layer	Java Spring Boot	Framework for creating the core backend business logic and microservices, ensuring robustness and scalability.
Identity & Access	Keycloak	Manages identity and access control, handling user federation, Role-Based Access Control (RBAC), Multi-Factor Authentication (2FA/MFA), and Single Sign-On (SSO).
Web & Proxy Layer	Nginx	Serves as the primary web server, reverse proxy, load balancer, and Web Application Firewall (WAF) to manage and secure inbound traffic.
Data Layer	PostgreSQL	A powerful, open-source object-relational database system for storing structured application data, including user records, application details, and transaction logs.
	MinIO	An S3-compatible object storage solution for scalable and secure management of unstructured data, such as uploaded documents (PDFs, images, test reports).
	Milvus	A highly scalable, cloud-native vector database optimized for similarity searches, supporting the AI workloads for features like document analysis and fraud detection.
AI Service Layer	Python FastAPI	A high-performance Python framework for building the AI backend logic and serving RESTful APIs for AI-powered features.

	LangChain / LangGraph	Frameworks used to orchestrate complex Large Language Model (LLM) workflows and build intelligent agents for tasks like document validation and chatbot responses.
	vLLM / NVIDIA NIM	High-throughput and efficient inference engines for serving the AI Large Language Models, ensuring fast and reliable AI performance.
API & Integration	Apache APISIX	A dynamic, real-time, high-performance API Gateway used for traffic control, security policy enforcement, and comprehensive API observability.
	IBM webMethods	Functions as the Enterprise Service Bus (ESB) and Integration Hub to facilitate secure, reliable, and managed bidirectional data exchanges with external systems (SIRIM, RMCD) and internal MCMC systems (SIFS, AAD).
Containerization	Docker	The platform for developing, shipping, and running applications in isolated containers, ensuring consistency across all environments.
Orchestration	Rancher	An enterprise-grade platform for managing Kubernetes clusters, used for orchestrating the deployment, scaling, and management of containerized applications.

3.2. Deployment Architecture

The NCEF system is deployed across three distinct environments: Development, UAT/Staging, and Production.

The infrastructure is based on virtual machines running Ubuntu 24.04.1 LTS, with containerized application components managed by Rancher (Kubernetes). The production environment is designed for high availability, with active-active and active-passive configurations for critical components.

Below is a high-level diagram illustrating the production deployment architecture.

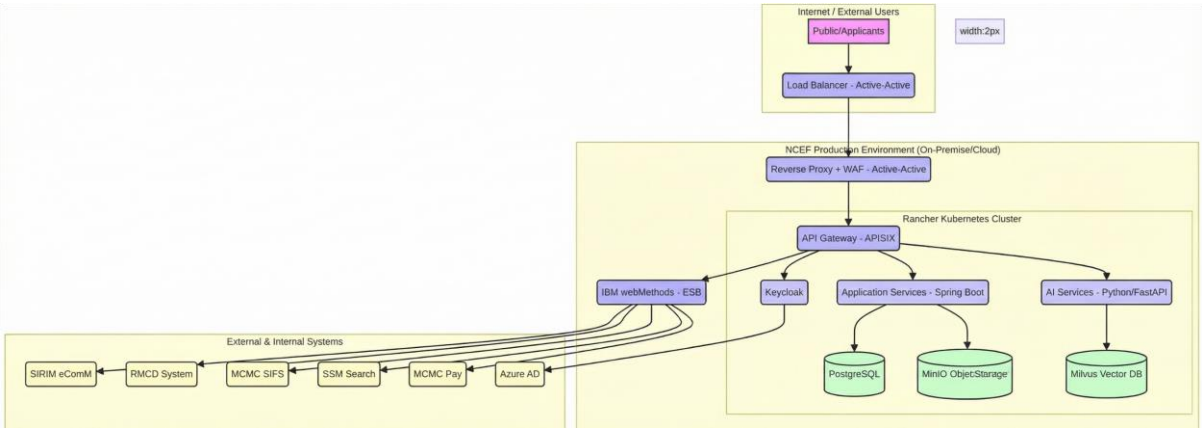


Figure: Deployment Architecture Diagram

3.3. Logical Architecture

The system is designed with a layered logical architecture to ensure a clear separation of concerns, which enhances maintainability, scalability, and security. Each layer has a distinct responsibility.

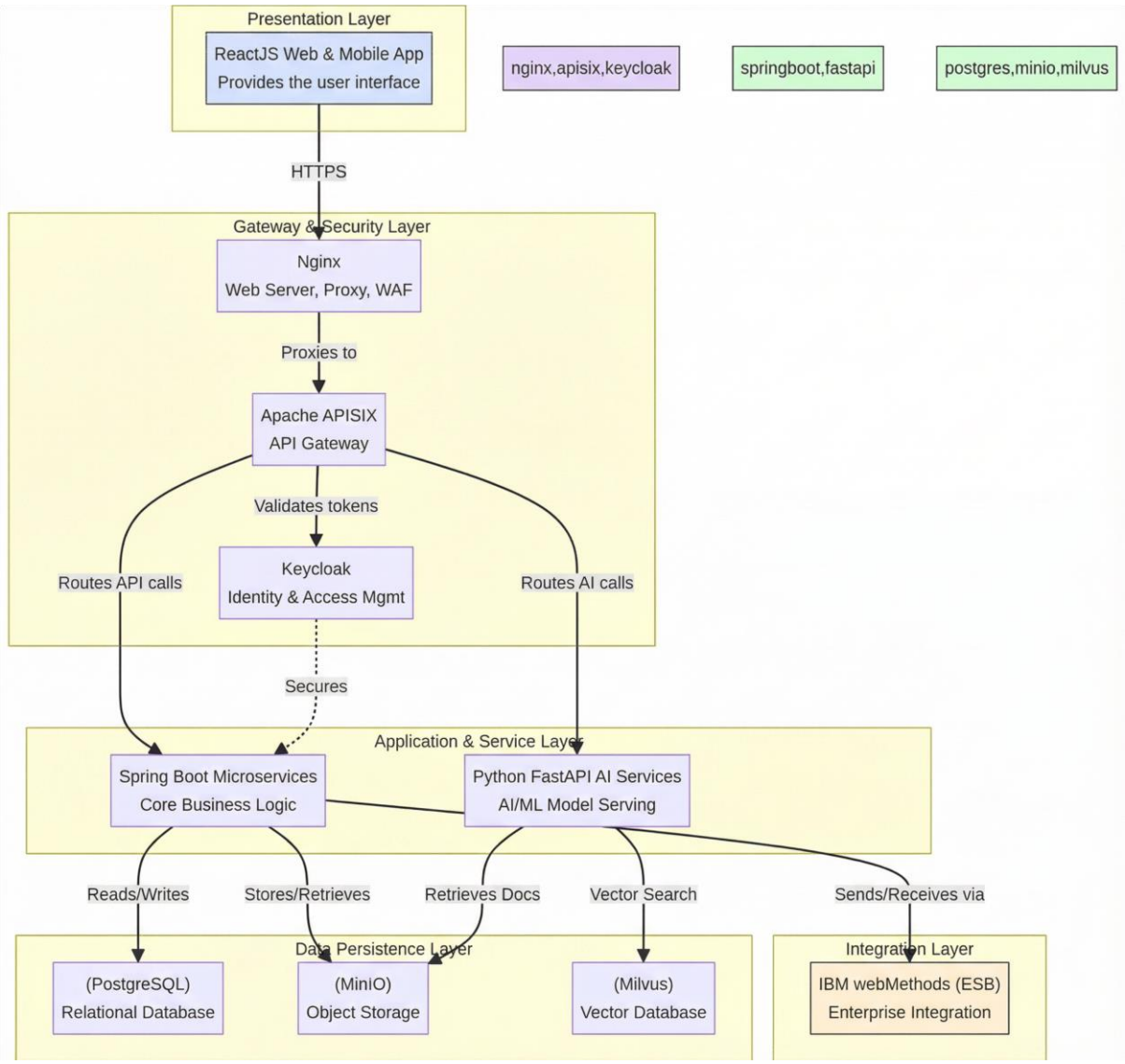


Figure: Logical Architecture Diagram

3.4. Integration Architecture

The NCEF system acts as a central hub, requiring seamless integration with multiple external and internal systems. The integration architecture is designed around an Enterprise Service Bus (ESB) pattern, using IBM webMethods as the central integration hub to ensure secure, reliable, and manageable data flows.

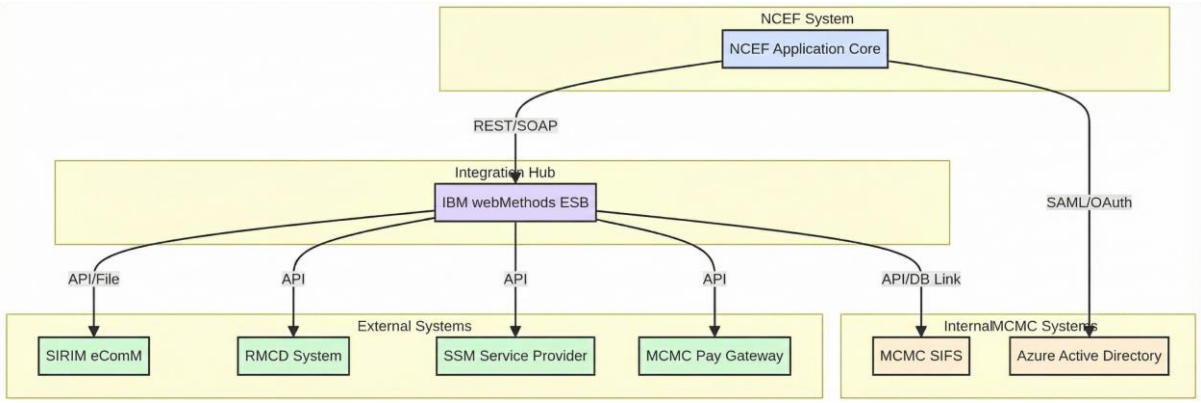


Figure: Integration Architecture Diagram

Integration Points:

- **SIRIM eComM:** For validating Certificate of Conformity (CoC) details and migrating historical data. Validation for Communications Equipment under Scheme A, verification of information for Schemes A, B, C, and special approval for importation, and data migration to the current platform.
- **RMCD MyOGA System:** For processing import permit applications.
- **SSM Search:** For real-time validation of company registration details during applicant onboarding.
- **MCMC Pay:** For processing all financial transactions, including application, renewal, and other fees.
- **MCMC SIFS:** For synchronizing all transaction data for financial reporting and reconciliation.
- **Azure Active Directory (AAD):** For enabling Single Sign-On (SSO) for internal MCMC users.

4. Data Design

This section describes the data architecture of the NCEF system, including the logical data model and a dictionary of key data entities. The data design ensures that all information required to support the functional requirements is structured efficiently, securely, and in a scalable manner.

4.1. Data Model (Entity-Relationship Diagram)

The following diagram illustrates the high-level relationships between the core entities of the NCEF system. The primary database will be PostgreSQL for structured data, while MinIO will handle unstructured document storage.

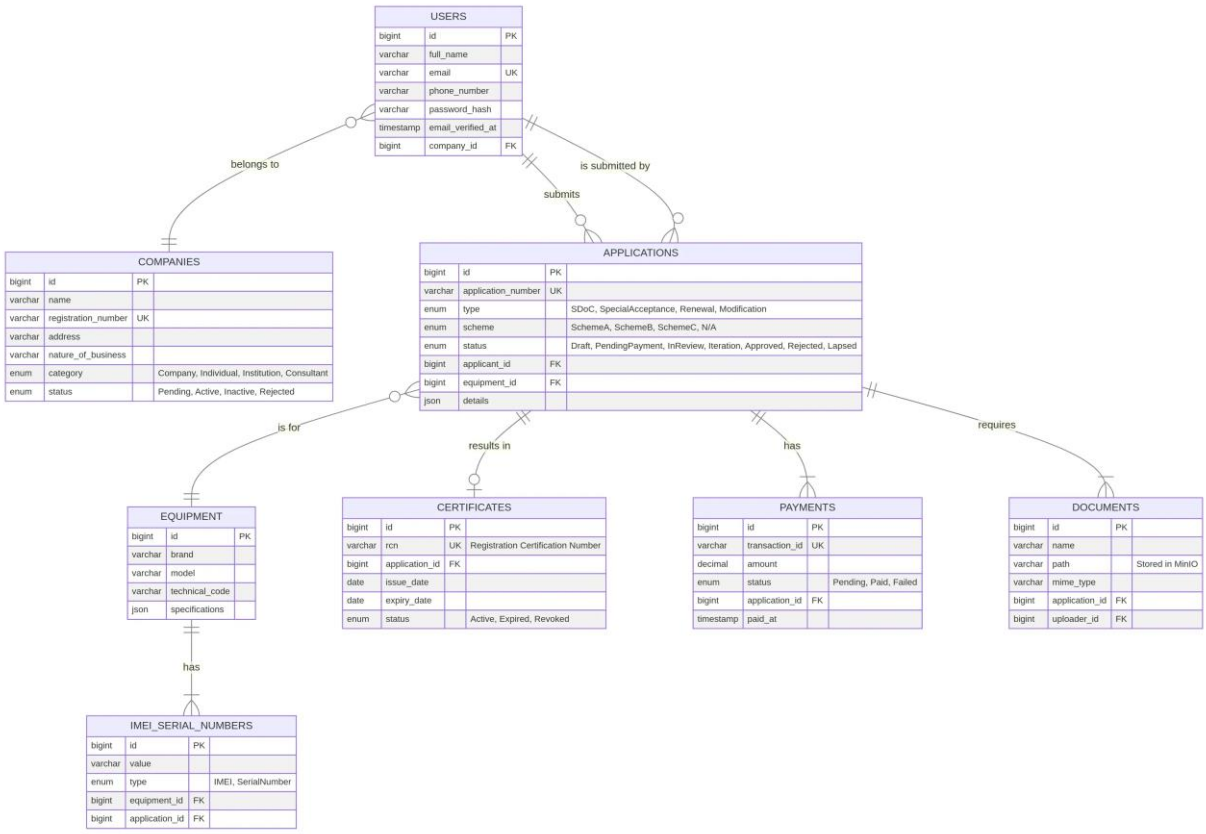


Figure: Initial Entity-Relationship Diagram

4.2. Data Dictionary

This table describes the key data entities and their primary attributes.

Entity	Description	Key Attributes
Users	Represents an individual with access to the system, either an external applicant or an internal MCMC officer.	id, full_name, email, phone_number, password_hash, company_id (FK to Companies)
Companies	Represents a registered entity (Company, Individual, Institution, or Consultant) that submits applications.	id, name, registration_number, category, status (of registration)
Applications	The central entity representing any submission into the system, such as an SDoC, Special Approval, or Renewal.	id, application_number, type, scheme, status, applicant_id (FK to Users), equipment_id (FK to Equipment)
Equipment	Represents a specific model of communications equipment being registered.	id, brand, model, technical_code, specifications (JSONB for technical details)
Certificates	The RCN is a unique number that serves as the identifier for a product registration. A certificate is issued containing all relevant information pertaining to the registered product. In the process flowcharts, the 'Issuance of RCN' refers to the generation of the RCN and the issuance of the corresponding certificate. This applies to all schemes.	id, rcn (Registration Certification Number), application_id (FK to Applications), issue_date, expiry_date
Payments	Records a financial transaction associated with an application, processed via MCMC Pay.	id, transaction_id, amount, status, application_id (FK to Applications)
Documents	Represents a file uploaded to the system as part of an application	id, name, path (MinIO object key), application_id (FK to Applications)

	(e.g., test report, CoC). The file itself is stored in MinIO.	
IMEI_Serial_Numbers	Stores the unique identifiers (IMEI or Serial Number) for registered equipment units.	id, value, type, equipment_id (FK to Equipment)

5. Component Design

This section provides a detailed design for each of the major components (modules) of the NCEF system. Each subsection describes the module's purpose, key functionalities, interfaces, and data requirements, based on the URS [1].

5.1. User Management Module

5.1.1. Purpose

The User Management Module is the foundation for securing the NCEF system. It handles the entire lifecycle of user accounts, including registration, authentication, authorization, and profile management for all internal (MCMC) and external (Applicant) users. It leverages Keycloak for robust identity and access management.

5.1.2. Key Functionalities

User Classification: The system classifies users into two primary groups:

- **Internal Users (MCMC):** System Administrator, Officer in Charge (OIC), Recommender, Verifier, Approver.
- **External Users (Applicants):** Category A (Company), B (Individual), C (Institution), D (Consultant).

External User Registration: A two-layer verification process for new external users:

1. **Account Creation (Layer 1):** User provides basic details (Name, Email, Phone) and creates a password. An email verification link is sent.
2. **Profile Registration (Layer 2):** After email verification, the user logs in to complete their profile by selecting a category and submitting detailed information and required documents (e.g., Business Registration for companies).
3. **MCMC Acceptance:** The submitted profile undergoes a review and acceptance workflow by MCMC officers before the account is fully activated.

Internal User Authentication (SSO): Internal MCMC users will authenticate via Single Sign-On (SSO) integrated with MCMC's Azure Active Directory (AAD). Keycloak will act as the identity broker, federating with AAD.

External User Authentication (2FA): External users will log in using their email and password, with a Two-Factor Authentication (2FA) step. The first login after registration will use an OTP sent to the user's email to enforce verification. As a standard practice, users will time out after 30-minutes of inactivity, which ends their session. User may access the platform from multiple devices and tabs at the same time.

Role-Based Access Control (RBAC): Keycloak will manage roles and permissions. A detailed User Access Matrix (as defined in URS Appendix A) will be configured in Keycloak to control access to different modules and functionalities.

Profile Management: All users can view and update their profile information after login.

Principal & Consultant Management: The system allows a primary supplier (Company) to register and link subsidiary Principals or third-party Consultants to their account, with the ability to manage (add/remove) these associations, but Principals and Consultants remain active as long as their account is renewed. Existing selection of Principals and Consultants are available via dropdown.

5.1.3. Interfaces

UI: ReactJS components for registration, login, profile management, and user administration screens.

Backend API: Spring Boot provides RESTful endpoints for all user management functions (e.g., /api/users/register, /api/users/profile).

Keycloak: The module integrates directly with Keycloak's Admin API for user creation and role mapping, and with its authentication endpoints for login flows.

Azure Active Directory (AAD): Keycloak is configured to use AAD as an external identity provider for SSO.

SSM Search: During company registration, the system calls the SSM Search API via the ESB to validate business registration details.

5.1.4. Data Requirements

PostgreSQL: USERS, COMPANIES tables.

MinIO: Stores uploaded registration documents (e.g., copy of NRIC, Business Registration forms).

Keycloak DB: Stores user credentials, roles, sessions, and realm configurations.

5.2. Equipment Registration Module (SDoC)

5.2.1. Purpose

The Equipment Registration Module is the central component of the NCEF system, designed to manage the Supplier's Declaration of Conformity (SDoC) process. It provides distinct, risk-based workflows for registering communications equipment, categorized as Scheme A (High Risk), Scheme B (Medium Risk), and Scheme C (Low Risk).

5.2.2. Key Functionalities

SDoC Form Submission:

Applicants initiate registration by completing a comprehensive SDoC form. The form captures equipment details, technical specifications, test reports, and labelling information as specified in the URS (Section 4.2.4).

Scheme A Workflow (High Risk - SDoC with Certification):

1. Applicant submits the SDoC form and all required documents (e.g., Certificate of Conformity from a recognized body).
2. Applicant proceeds to payment via the MCMC Pay gateway.
3. Upon payment confirmation, the system assigns the application to an MCMC Officer in Charge (OIC).
4. The AI engine performs an initial validation of documents and data, generating a summary report for the OIC.
5. The OIC conducts a detailed manual review of the application, supporting documents, and the AI summary.
6. Based on the review, the OIC can accept the application, not accept it, or send it back to the applicant for modification (iteration).
7. Upon approval, a Registration Certification Number (RCN) and a digital certificate are issued.

Scheme B Workflow (Medium Risk - SDoC with Verification):

- The workflow mirrors Scheme A, with the AI engine providing significant decision support to the OIC. A key feature in this workflow is the ability for the officer to **reclassify** the application to either Scheme A (if higher risk is determined) or Scheme C (if lower risk is determined), which may require the applicant to provide additional or different information.

Scheme C Workflow (Low Risk - AI Auto-Acceptance):

- The process is identical to Scheme B up to the AI validation step.
- The AI engine analyzes the application and generates a confidence score based on pre-defined criteria (e.g., SSM validation, document completeness, supplier history, as per URS 4.2.3).
- If the confidence score is above the configured threshold (e.g., 90%), the application is **auto-accepted**, and an RCN is issued without manual intervention.
- If the score is below the threshold, the application is routed to an OIC for manual review, similar to the Scheme B workflow.

Digital Signature: All applications must be concluded with a formal declaration by the applicant's authorized officer, captured as a digital acceptance in the system.

5.2.3. Interfaces

UI: ReactJS components for the multi-part SDoC form, document upload interface, and application status tracking dashboard.

Backend API (Spring Boot): Endpoints to create and manage applications (e.g., /api/applications/sdoc), handle state transitions, and trigger notifications.

AI Service (Python/FastAPI): The backend will call the AI service API to submit documents for ICR, data extraction, and to receive the confidence score and validation summary.

Payment Gateway (MCMC Pay): The system redirects the user to MCMC Pay for payment and receives a callback confirmation via the ESB.

5.2.4. Data Requirements

PostgreSQL: APPLICATIONS, EQUIPMENT, CERTIFICATES, DOCUMENTS tables are central to this module.

MinIO: Stores all uploaded technical documents, test reports, and Certificates of Conformity.

Milvus: May store vector embeddings of technical documents to assist the AI in comparing specifications and detecting inconsistencies.

5.3. Special Approval Module

5.3.1. Purpose

The Special Approval Module provides a specialized workflow for importing communications equipment for specific, non-commercial purposes, such as R&D, market surveys, demonstrations, or personal use. The module implements distinct processes based on the equipment's risk classification: Low/Medium Risk, High Risk, and Prohibited.

5.3.2. Key Functionalities

Application Submission: Applicants initiate a request by selecting the purpose of usage and providing details about the equipment, usage location, and time period. They must also upload supporting documents like proposal papers or letters of undertaking.

Low/Medium Risk Workflow:

1. **Application Submission:** The applicant fills in the required information, including the purpose of usage, usage details (location, time period), and detailed equipment information.
2. **Document Upload:** The applicant uploads all necessary supporting documents, such as a proposal paper, technical data sheets, and letters of undertaking.
    a. This includes submitting IMEI or Serial Number, whichever preferred in a bulk upload format
3. **Notice for Payment:** The system generates a notice for payment based on the equipment type, quantity, and corresponding registered serial/IMEI numbers. The applicant completes the payment via MCMC Pay.
4. **Officer Assignment & AI Evaluation:** Upon payment, the application is assigned to an officer, and the AI engine evaluates the submitted documents.
5. **OIC Verification:** An OIC reviews the application and the AI summary. If modifications are needed, the application is returned to the applicant for iteration.
6. **Acceptance and Certificate Issuance:** Upon successful verification, the OIC accepts the application, and the system generates and issues a Special Approval Certificate with a unique ID (SA-MMYY-XXXXX).

High Risk Workflow:

This process is identical to the Low/Medium risk workflow but involves a different (higher) fee structure. The final output is a **Special Approval Certificate**.

1. After submission, the system generates a notice for payment based on the equipment type and quantity.

2. Upon payment confirmation via MCMC Pay, the application is assigned to an **Recommender** OIC.
3. The AI engine evaluates the submitted documents and provides a summary to the officer.
4. The **Recommender** OIC reviews the application and can escalate to **Approver** by accepting the submission or send it for iteration, both requiring justifications.
5. Approver will then accept, not accept or rev
6. Upon approval, a **Special Approval Certificate** is issued.

Prohibited Equipment Workflow:

1. This workflow is for equipment that is normally prohibited and requires stringent oversight.
2. A critical step involves an **offline acceptance process** (e.g., a physical meeting) where a decision is formally minuted.
3. A designated MCMC officer (Recommender) uploads the official decision document (e.g., meeting minutes) to the system.
4. The application is then routed through a mandatory, multi-level digital approval workflow involving a **Recommender**, a **Verifier**, and a final **Approver**
5. Only **Approver** shall be able to *Reject*, whereas **Recommender** and **Verifier** shall only be able to escalate by approving the request, and provide justification while highlighting potential approval/rejection for Approver to review.
6. Once all levels of digital approval are complete, the system issues the **Special Approval Certificate**.

5.3.3. Interfaces

UI: ReactJS components for the Special Approval application form, document upload, and status tracking.

Backend API (Spring Boot): Endpoints to manage special approval requests (e.g., /api/applications/special-approval) and the multi-level approval chain for prohibited equipment.

AI Service (Python/FastAPI): Used for document evaluation in the Low/Medium and High Risk workflows.

Payment Gateway (MCMC Pay): Integration for fee processing.

5.3.4. Data Requirements

PostgreSQL: The APPLICATIONS table will store these requests with a type of 'SpecialApproval'. The details JSONB column will store purpose-specific information. A separate table may be required to manage the multi-level approval state for prohibited equipment.

©Infomina Berhad

MinIO: Stores all uploaded supporting documents, including proposal papers and the offline acceptance minutes for prohibited equipment.

5.4. Renewal Module

5.4.1. Purpose

The Renewal Module automates and manages the renewal processes for both applicant registrations (Supplier, Principal, Consultant) and equipment certifications (SDoC). Its primary goal is to ensure continuous compliance by preventing lapses in registration and certification validity through timely notifications and a streamlined renewal process.

5.4.2. Key Functionalities

Automated Renewal Notifications: The system will automatically send email and in-app notifications to applicants in advance of their registration or certificate expiry dates. The notification schedule will be configurable (e.g., 90, 60, and 30 days before expiry).

Supplier/Principal/Consultant Renewal:

1. Applicants can initiate the renewal from their dashboard or via a link in the notification email.
2. The system prompts the user to review and update their profile and company information.
3. The renewal application undergoes a simplified acceptance workflow, where the AI will review for any discrepancies and approve if the risk score is above 90%; else the renewal shall be escalated to OIC.
4. A notice for payment is generated based on the selected renewal period (1 to 5 years).
5. Upon successful payment via MCMC Pay, the user's account status is updated, and the validity period is extended.

Equipment Registration Renewal (SDoC):

1. Applicants can select expired or soon-to-expire certificates from their dashboard to initiate renewal.
2. The renewal process largely mirrors the original application workflow but is streamlined:
    a. **Scheme A:** Requires the applicant to upload an updated Certificate of Conformity (CoC).
    b. **Scheme B & C:** Do not require document updates unless there are changes to the underlying technical standards.
3. The AI-assisted compliance check re-validates the equipment against the latest technical standards.
4. Upon successful validation and payment, the certificate's validity is extended, and a new digital certificate may be issued.

5.4.3. Interfaces

UI: ReactJS components for the renewal application forms and dashboard alerts.

Backend API (Spring Boot): Endpoints to handle renewal submissions (e.g., /api/applications/renewal), calculate fees, and update expiry dates.

Notification Service: A dedicated service (or part of the core application) triggered by a scheduled job (cron) to check for upcoming expiries and send notifications.

Payment Gateway (MCMC Pay): Integration for processing renewal fees.

5.4.4. Data Requirements

PostgreSQL: The APPLICATIONS table will store renewal requests. The COMPANIES and CERTIFICATES tables will have their expiry_date and status fields updated upon successful renewal.

5.5. IMEI/Serial Number Registration Module

5.5.1. Purpose

This module provides the functionality to register unique equipment identifiers—International Mobile Equipment Identity (IMEI) for cellular devices and Serial Numbers for non-cellular equipment. This is a critical function for tracking individual devices for post-market surveillance and customs clearance.

5.5.2. Key Functionalities

Identifier Registration: Applicants can submit IMEI or Serial Numbers to be associated with a specific registered equipment model (identified by an RCN).

Bulk Upload: To handle large quantities efficiently, the system supports the bulk upload of identifiers via a file import feature (CSV format). The file template will be downloadable from the portal.

Duplication Validation: The system will perform validation to ensure that each submitted identifier is unique within the NCEF database for its type.

- **IMEI:** Validated for uniqueness across all registered IMEI numbers.
- **Serial Number:** Validated for uniqueness for the same supplier and RCN.

Fee Calculation: The system automatically calculates the total fee based on the quantity of identifiers submitted (RM 0.50 per IMEI, RM 0.15 per Serial Number) and generates a notice for payment.

5.5.3. Interfaces

UI: ReactJS components for the bulk upload interface, including file selection, validation preview, and error reporting.

Backend API (Spring Boot): An endpoint (e.g., /api/identifiers/register) to handle the file upload, parse the data, perform validation against the database, and create payment records.

5.5.4. Data Requirements

PostgreSQL: The IMEI_SERIAL_NUMBERS table is the primary entity for this module, storing the identifier value, type, and its link to the parent equipment registration.

5.6. Modification of Registration Module

5.6.1. Purpose

This module allows applicants to submit requests to modify the details of their existing equipment registrations. It provides a formal process for informing MCMC of changes and ensures that a complete audit trail of all modifications is maintained.

5.6.2. Key Functionalities

Modification Request Initiation: An applicant initiates a request by entering a valid Registration Certification Number (RCN). The system validates the RCN and retrieves the current equipment information.

Modification Categorization: The applicant categorizes the nature of the modification by selecting one or more of the following:

- **Major:** Significant changes that may affect the equipment's conformity (e.g., changes to radio frequency components). Selecting this option redirects the applicant to file a new registration application.
- **Minor:** Small changes that do not affect conformity (e.g., color, enclosure material).
- **Others:** Administrative changes (e.g., updating the applicant's contact person).

Acceptance Workflow: For **Minor** and **Others** modifications, the request is assigned to a designated MCMC officer for review and verification. The officer can accept or not accept the modification request.

Version Control: The system will maintain a version history for every registered equipment, creating a new version for each accepted modification. This ensures a complete and traceable audit trail of all changes over the certificate's lifecycle.

5.6.3. Interfaces

UI: ReactJS components for the modification request form.

Backend API (Spring Boot): Endpoints to handle modification submissions (e.g., /api/applications/modification) and manage the officer review workflow.

5.6.4. Data Requirements

PostgreSQL: The APPLICATIONS table will store modification requests. The EQUIPMENT and CERTIFICATES tables will be designed to support versioning, either through a history table or by maintaining version numbers and status flags on the main records.

©Infomina Berhad

5.7. Importation Module

5.7.1. Purpose

This module streamlines the process for obtaining import permits for communications equipment. It achieves this by integrating with the Royal Malaysian Customs Department (RMCD) system, allowing applicants to initiate the permit process from within the NCEF portal.

5.7.2. Key Functionalities

Import Permit Application: Applicants can initiate an import permit application from their dashboard, providing comprehensive details including Trader information (registered with RMCD), Consignor/Consignee details, Agent details, and logistics information (mode of transport, country of origin, etc.).

Integration with RMCD MyOGA System: The NCEF system will integrate with the RMCD's electronic system via the IBM webMethods ESB. The application data is securely transmitted to the RMCD MyOGA System.

Redirection for Payment: To finalize the process, the applicant is redirected from the NCEF portal to the RMCD MyOGA System to complete the necessary payments. All fees are determined and collected directly by RMCD.

Certificate of Acceptance (CoA): Upon successful payment confirmation within the RMCD MyOGA System, the system generates the Certificate of Acceptance (CoA), which specifies the total allowed import quantities.

5.7.3. Interfaces

UI: ReactJS components for the import permit application form.

Backend API (Spring Boot): An endpoint to gather application data and forward it to the ESB for transmission to RMCD MyOGA System.

Integration Hub (IBM webMethods): Manages the secure API-based communication between the NCEF system and the RMCD MyOGA System.

5.7.4. Data Requirements

PostgreSQL: The APPLICATIONS table may store a record of the import permit request initiation. However, the primary data store for the permit itself resides within the RMCD MyOGA System.

5.8. Post-Market Surveillance (PMS) Module

5.8.1. Purpose

The Post-Market Surveillance (PMS) Module supports MCMC in monitoring the ongoing compliance of registered communications equipment and suppliers after they have entered the market. It enables end-to-end audit management, product sampling, and compliance verification to ensure strict adherence to MCMC regulatory standards.

5.8.2. Key Functionalities

Audit Proposal Generation (AI-Assisted): The system utilizes the AI Risk Scoring Engine to automatically generate proposed audit lists for suppliers and registered equipment. The selection is based on configurable criteria including equipment risk classification (Scheme A, B, C), supplier compliance history, registration volume, random sampling, and customized parameters defined by officers.

Product Sampling Proposal: Officers can define a specific sampling percentage for registered equipment, prompting the system to automatically generate a proposed product sampling list for review and confirmation.

Supplier Notification: Automatically generates and dispatches notifications (via email and system alerts) to suppliers selected for audits or product sampling, with templates that officers can review and edit before sending.

Audit Findings & Document Updates: Provides a dedicated interface for officers to record verification results, capture observations, and upload supporting audit documents.

Audit History & Non-Compliance Records: Maintains an immutable history of all audit activities and outcomes. It automatically flags and records suppliers and equipment identified as non-compliant, directly feeding this data back into the AI Risk Scoring Engine to influence future audit selections and auto-acceptance thresholds.

5.8.3. Interfaces

UI: ReactJS components for the Officer Dashboard to configure audit parameters, review AI-proposed lists, and input audit findings.

Backend API (Spring Boot): Endpoints to manage audit lifecycles and sampling plans (e.g., /api/pms/audits, /api/pms/sampling).

AI Service (Python/FastAPI): The backend triggers the AI service to analyze historical data, complaints, and supplier profiles to generate the risk-based audit proposals.

Notification Service: Triggered by the backend to dispatch alerts to the Applicant Dashboard and via email.

5.8.4. Data Requirements

PostgreSQL: New AUDITS, PRODUCT_SAMPLING, and NON_COMPLIANCE tables linked to the existing COMPANIES and EQUIPMENT tables.

MinIO: Securely stores uploaded physical audit evidence, test reports, and scanned documentation.

5.9. Complaint & Surveillance Module

5.9.1. Purpose

This module enables MCMC officers to centralize, record, and manage public complaints and market intelligence related to communications equipment currently active in the market.

5.9.2. Key Functionalities

Complaint Submission: Allows officers to capture data from public complaints. Data captured includes supplier name, brand/model, IMEI/Serial Number, issue description, and source.

Complaint Tracking: Provides a dashboard view for authorized officers to monitor the lifecycle of complaints, current status, and related equipment/supplier links.

5.9.3. Interfaces

UI: ReactJS forms for manual intelligence entry and dashboard tracking views.

Backend API (Spring Boot): Endpoints for creating and querying complaint records (e.g., /api/intelligence/complaints).

5.9.4. Data Requirements

PostgreSQL: New COMPLAINTS table mapping directly to EQUIPMENT and COMPANIES for relational tracking.

5.10. Compliance Status Management Module

5.10.1. Purpose

To provide authorized MCMC officers with the direct capability to govern and update the overarching compliance status of registered equipment and suppliers based on audit findings, intelligence, or enforcement decisions.

5.10.2. Key Functionalities

- Compliance Status Controls: Officers can update the status of equipment or suppliers to categories such as Active, Under Surveillance, Suspended, or Cancelled.
- Status Recording & Audit Trail: Enforces strict logging for any status change, capturing the reason for the change, the officer responsible, and the exact date/time.
- System-Wide Propagation: Status updates immediately propagate throughout the system. For instance, a "Suspended" supplier status will restrict their ability to submit new SDoCs, and a "Cancelled" equipment status will instantly reflect on the Public Search Portal.

5.10.3. Interfaces

- UI: ReactJS administrative controls located within the MCMC Officer profile management and equipment detail screens.
- Backend API (Spring Boot): Secure endpoints managing status state transitions (e.g., /api/compliance/status-update).

5.10.4. Data Requirements

PostgreSQL: Additions of compliance_status columns to the COMPANIES and CERTIFICATES tables. A new COMPLIANCE_STATUS_HISTORY table to maintain the immutable audit trail.

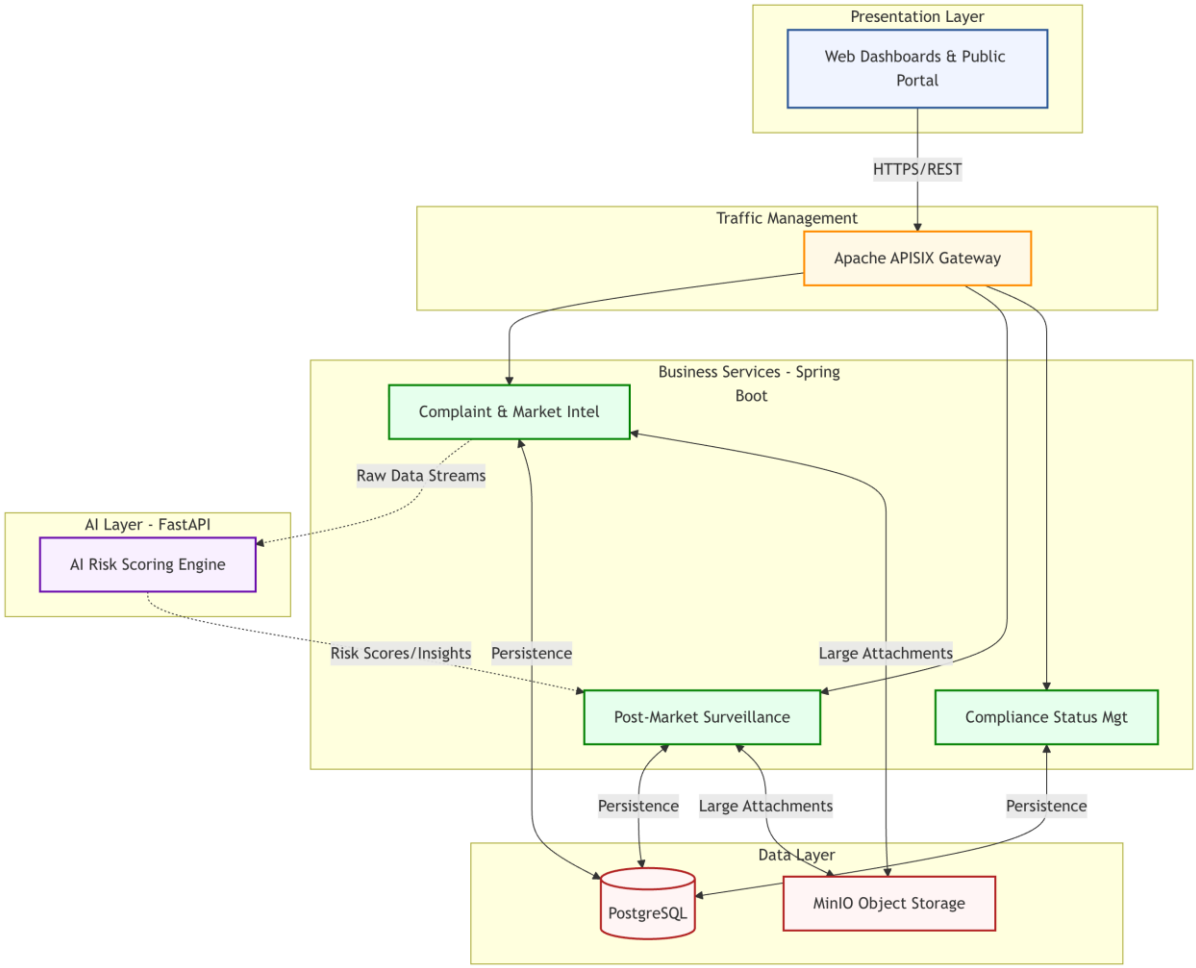


Figure: System Architecture for Post-Market Flow

5.10. Public Module

5.10.1. Purpose

The Public Module serves as the public-facing gateway to the NCEF system. It is designed to provide transparent access to information for the general public, enhance consumer confidence, and serve as a central repository for all official NCEF-related guidelines and announcements.

5.10.2. Key Functionalities

Public Landing Page: A professional, user-friendly, and bilingual (English and Malay) landing page that serves as the main entry point. It will feature the latest announcements, regulatory updates, and news from MCMC.

Equipment Certification Search & Verification: A powerful and easy-to-use search engine that allows any member of the public to verify the registration status of a communication device. The search function will support multiple parameters, including Brand, Model, IMEI/Serial Number, Supplier ID, and Registration Certification Number (RCN).

Document Repository: Provides public access to a repository of all relevant NCEF guidelines, procedures, applicable technical standards, and other reference documents.

Application Process Overview: A dedicated section featuring a step-by-step overview of the registration application process, including indicative processing timelines and a clear, easy-to-understand fee structure.

AI Chatbot: An AI-powered chatbot will be available on the portal to provide instant answers to common inquiries related to the NCEF system, registration processes, and fees.

FAQ Section: A comprehensive and searchable Frequently Asked Questions (FAQ) section to address common queries.

Contact Information: Displays official contact details for NCEF-related inquiries.

5.10.3. Interfaces

UI: ReactJS components for the public landing page, search interface, document repository, and FAQ section.

Backend API (Spring Boot): Publicly accessible (but secured) endpoints to power the search functionality (e.g., /api/public/search).

AI Service (Python/FastAPI): The AI chatbot on the UI will interact with a dedicated endpoint on the AI service.

5.10.4. Data Requirements

PostgreSQL: The search functionality will query a read-optimized view of the EQUIPMENT and CERTIFICATES tables. The FAQ and document repository content will be stored in dedicated tables managed by MCMC content managers.

Milvus: The AI chatbot will leverage the vector database to find relevant information from the knowledge base (guidelines, FAQs) to answer user queries.

5.11. Mobile Application

5.11.1. Purpose

To provide users with on-the-go access to key NCEF functionalities and to serve as a channel for important announcements from MCMC. The application will be developed using the Flutter framework, ensuring a consistent user experience across all platforms (Android, iOS, Huawei) while streamlining development and maintenance.

5.11.2. Key Functionalities

The mobile application will provide a focused set of features optimized for smaller screens:

Public Equipment Search: A mobile-friendly version of the public search and verification feature, allowing users to quickly check the registration status of a device.

Push Notifications: The application will support push notifications, which will be used exclusively for broadcasting important announcements, regulatory updates, and public notices from MCMC.

Responsive UI: The application will feature a clean, intuitive, and responsive user interface that adapts seamlessly to various screen sizes and orientations.

5.11.3. Interfaces

UI: The application will be built with Flutter, with the design elements referencing the web portal as well as the native design system of the device operating system.

Backend API (Spring Boot): It will consume the same public API endpoints as the web-based Public Module for search functionality.

Push Notification Service: It will integrate with a push notification service (like Firebase Cloud Messaging or a self-hosted solution) to receive and display notifications sent by MCMC administrators.

5.11.4. Data Requirements

The mobile application is primarily a client-side application and does not have its own backend data store. It relies on the main NCEF database via the public API.

5.12. Payment Integration Module

5.12.1. Purpose

This module is responsible for handling all financial transactions within the NCEF system in a secure, reliable, and integrated manner. It ensures that all fees for registration, renewal, and other services are processed correctly and reconciled with MCMC's financial systems.

5.12.2. Key Functionalities

MCMC Pay Integration: The system will be seamlessly integrated with MCMC Pay, the official payment gateway for MCMC. All payments will be processed through this gateway.

Fee Structure & Notice for Payment: The system will automatically generate notice for payment with the quotation based on a configurable fee structure. This includes different fees for various schemes, risk levels, registration types, and quantities (e.g., for IMEI/Serial Number registration).

Payment Confirmation: Upon successful payment completion on the MCMC Pay portal, the NCEF system will receive a real-time confirmation callback. This will trigger an update to the corresponding application's status (e.g., from "Pending Payment" to "In Review").

Digital Receipt Issuance: The system will automatically issue a digital receipt to the applicant via email upon successful payment confirmation, using a standard template from NCEF platform.

Financial Reconciliation (SIFS Integration): All transaction data will be consolidated and synchronized with MCMC's Integrated Financial System (SIFS) via the ESB to ensure accurate financial reporting and reconciliation.

5.12.3. Interfaces

UI: ReactJS components will display the Notice for Payment details and provide a button to redirect the user to the MCMC Pay portal.

Backend API (Spring Boot): Endpoints to generate Notice for Payments, and a dedicated callback endpoint to receive payment status updates from the integration hub.

Integration Hub (IBM webMethods): Manages the communication flow between the NCEF system, MCMC Pay, and MCMC SIFS.

5.12.4. Data Requirements

PostgreSQL: The PAYMENTS table will store records of all financial transactions, including the transaction ID from MCMC Pay, amount, status, and a foreign key to the APPLICATIONS table.

5.13. AI-Enabled Features

5.13.1. Purpose

This component encapsulates the Artificial Intelligence (AI) capabilities of the NCEF system. It is designed to enhance operational efficiency, improve accuracy in application processing, and provide intelligent decision support for MCMC officers. The AI services are built on a dedicated Python-based microservices stack.

5.13.2. Key Functionalities

AI Document Validation (ICR): The AI engine will perform Intelligent Character Recognition (ICR) to extract text and structured data from uploaded documents (e.g., PDFs, scanned images of test reports). It will check for document completeness and cross-reference information (e.g., model numbers) across different documents for consistency.

Risk Scoring & Assessment: For each application, the AI will generate a risk score based on a combination of factors, including equipment type, applicant history, and the results of the document validation. This score helps officers prioritize reviews.

AI-Assisted Decision Support (Scheme A, B and C): The AI provides a clear recommendation (e.g., “Accept” or “Flag for Review”) to the OIC, along with a summary of its findings, highlighting potential compliance issues or inconsistencies.

Auto-Acceptance (Scheme C): For low-risk applications, the AI calculates a confidence score. If this score exceeds a configurable threshold (e.g., 90%), the application is automatically accepted without manual intervention.

Fraud Detection: The AI will be trained to identify patterns indicative of fraudulent activity, such as forged documents or suspicious application data, and flag these cases for immediate investigation.

AI Chatbot: An LLM-powered chatbot, available on the public portal, will use a Retrieval-Augmented Generation (RAG) approach. It will query the Milvus vector database (containing indexed guidelines and FAQs) to provide accurate, context-aware answers to user inquiries.

5.13.3. Interfaces

Backend API (Spring Boot): The main application will interact with the AI services via RESTful API calls for tasks like submitting a document for analysis or retrieving a risk score.

AI Service API (Python/FastAPI): Exposes endpoints for all AI functionalities (e.g., /api/ai/validate-document, /api/ai/get-risk-score, /api/ai/chat).

UI (ReactJS): The chatbot interface will directly communicate with the AI service API.

5.13.4. Data Requirements

MinIO: The AI service retrieves documents from MinIO for analysis.

PostgreSQL: The AI service accesses application and user data from the main database to inform its risk assessments.

Milvus: The vector database is the primary data source for the RAG-powered chatbot, storing embeddings of all knowledge base documents.

5.14. Dashboard & Reporting Module

5.14.1. Purpose

This module provides real-time insights and analytics into the NCEF system's operations for both internal MCMC officers and external applicants. It is designed to offer at-a-glance operational awareness, track Key Performance Indicators (KPIs), and generate detailed reports for compliance and strategic planning.

5.14.2. Key Functionalities

Internal Officer Dashboard: A comprehensive dashboard for MCMC officers displaying key metrics such as:

- Total application submission counts (daily, weekly, monthly).
- Breakdown of applications by Scheme, Special Approval type, and current status.
- Real-time SLA monitoring for application processing times.
- A personalized task list showing assigned applications and pending actions.

Applicant Dashboard: A personalized dashboard for external users, allowing them to:

- Track the real-time status of all their submitted applications.
- View a complete history of past and current applications.
- Receive and view system notifications and alerts.
- Initiate new applications, renewals, or modifications.

Analytics & Reporting: The system will be capable of generating a variety of detailed reports for MCMC management, including:

- Ability to export Master Report upon request, based on specific criterias (eg. Date, Time, types of applications, such as Supplier Registration, Equipment Registration, Special Approval, IMEI/Serial Number Registration, and other related application types)
- Compliance reports on registered equipment and accepted suppliers.
- Operational reports on officer workload and processing bottlenecks.
- Financial reports on payment transactions (to be reconciled with SIFS).
- A complete, immutable audit trail report for any user or application.
- Reports shall be available in .xlsx and .csv formats.

5.14.3. Interfaces

UI: ReactJS components for displaying charts, graphs, data tables, and personalized task lists.

Backend API (Spring Boot): Endpoints to provide aggregated data for the dashboards and to generate and export reports (e.g., in CSV or PDF format).

5.14.4. Data Requirements

PostgreSQL: The dashboard and reporting engine will query the main transactional database. For performance-intensive analytics, the system will use materialized views or a separate read-replica database to avoid impacting transactional performance.

5.15. Notification Module

5.15.1. Purpose

The Notification Module is responsible for keeping all stakeholders informed of key events, status changes, and required actions throughout the application lifecycle. It uses multiple channels to ensure timely and effective communication.

5.15.2. Key Functionalities

Multi-Channel Delivery: The system supports notifications via:

- **Email:** For all formal communications to both internal and external users (e.g., application submission confirmation, payment receipts, renewal reminders).
- **In-App/System Notifications:** Alerts displayed on the user's dashboard for important updates and pending actions.
- **Push Notifications:** For broadcasting important announcements from MCMC to users of the mobile application.

Event-Driven Triggers: Notifications are triggered by specific events in the system, such as application submission, status changes (e.g., Accepted, Request for Modification), payment confirmation, and upcoming expiry dates.

Configurable Templates: The content for all email and push notifications will be managed through configurable HTML/text templates. This allows System Administrators to modify notification wording and branding without requiring code changes.

5.15.3. Interfaces

Backend API (Spring Boot): The core application logic will trigger notification events.

Email Service: The backend will integrate with an email service provider (e.g., SMTP relay) to send emails.

Push Notification Service: The backend will integrate with a push service to send announcements to mobile clients.

5.15.4. Data Requirements

PostgreSQL: Notification templates will be stored in the database. A log of all sent notifications will also be maintained for auditing purposes.

5.16. Integration Requirements

5.16.1. Purpose

This component is responsible for managing all data exchanges between the NCEF platform and other enterprise systems. The design utilizes an Enterprise Service Bus (ESB) pattern, with IBM webMethods layer serving as the central Integration Hub, to ensure that all integrations are secure, reliable, manageable, and decoupled from the core application logic. SIRIM eComM and RMCD MyOGA System integration will be finalised once further clarification with each provider have been completed after initial discussion done on 23 February 2026.

5.16.2. Key Functionalities

SIRIM eComM System Integration:

- **Data Validation:** The NCEF application be communicating via IBM webMethod layer to send Certificate of Conformity (CoC), which will then call the SIRIM eComM API to validate the certificate's authenticity.
- **Data Migration:** The ESB will manage the three-phase migration of historical data from the legacy SIRIM system into the new NCEF database. This involves data transformation, cleansing, and loading processes orchestrated within IBM webMethods layer.

RMCD MyOGA System Integration: For import permit applications, the NCEF backend will send the application data to the ESB, which then forwards it to the RMCD MyOGA System's API.

SSM Service Provider Integration: During the company registration process, the NCEF backend will call the ESB via IBM webMethods with the company's registration number. The ESB will then connect to the SSM Service Provider's via SSM Search API to validate whether the business exists in SSM database or not. This information will be used to validate the company's existense under SSM, hence influencing the confidence score.

MCMC Pay Integration: The ESB alongside IBM webMethods will mediate all payment-related communication. It will pass payment requests to MCMC Pay and provide a secure callback endpoint to receive payment confirmation status, which it then forwards to the NCEF application.

MCMC SIFS Integration: The ESB will be responsible for synchronizing all confirmed payment transaction data from the NCEF database to MCMC's Integrated Financial System (SIFS), ensuring financial data consistency.

5.16.3. Interfaces

Backend API (Spring Boot): The core application will communicate with the ESB using well-defined RESTful APIs or SOAP web services for each integration point.

Integration Hub (IBM webMethods): Contains the core integration logic, including protocol transformation (e.g., REST to SOAP), data mapping, and routing to the various external and internal system APIs.

5.16.4. Data Requirements

This module is primarily for data in transit. The ESB will maintain detailed transaction logs for all integration activities for auditing and troubleshooting purposes.

5.17. Configuration & Settings

5.17.1. Purpose

This module provides a dedicated administrative interface for MCMC System Administrators to manage and configure various operational parameters of the NCEF platform. This allows for system behavior to be modified without requiring new code deployments, providing flexibility and reducing maintenance overhead.

5.17.2. Key Functionalities

System Parameter Configuration: Administrators can configure key system-wide parameters, such as application expiry timelines (e.g., number of days before a draft application is marked as lapsed) and payment deadlines.

Workflow Configuration: While the core logic is coded, key thresholds and assignments within workflows will be configurable. This includes setting the AI auto-acceptance confidence score for Scheme C and defining rules for officer assignment.

Officer Grouping & Assignment: Administrators can create and manage groups of officers and configure rules for automatic application assignment based on factors like equipment type or risk level. Administrators can appoint group leads which are able to override and apply manual re-assignment for leads, especially during unavailability of OIC.

Fee Structure Management: All fees within the system are configurable. Administrators can update the fees for different registration types, schemes, services (e.g., IMEI/Serial Number registration), and renewal periods.

Master Data Management: Administrators can manage the master lists used in application forms, such as the list of equipment types and applicable technical codes.

Officer Calendar Blocking: A feature allowing officers to block their calendars for off-duty days (leave, training). During a blocked period, the officer will be marked as unavailable and will not be eligible for new task assignments. For existing leads, it may be transferred to another OIC by the group leads.

5.17.3. Interfaces

UI: A dedicated section within the internal MCMC portal, built with ReactJS, providing administrative forms for managing all configurable parameters.

Backend API (Spring Boot): Secure endpoints accessible only to System Administrator roles for updating configuration values (e.g., /api/admin/settings).

5.17.4. Data Requirements

PostgreSQL: A dedicated settings or configuration table (or multiple tables) will be used to store all configurable parameters as key-value pairs or in a structured format.

5.18. Data Migration Strategy

5.18.1. Purpose

This component outlines the strategy for migrating all historical equipment registration data from the legacy system, previously managed by SIRIM, to the new NCEF platform. The primary objective is to ensure a seamless transition with high data integrity, completeness, and accuracy, making all historical records accessible within the new system from day one.

5.18.2. Key Functionalities & Phased Approach

The migration will be executed in three distinct phases, orchestrated via the IBM webMethods ESB, which will handle the Extract, Transform, and Load (ETL) processes.

Phase 1: Initial Full Load (Before UAT)

1. **Extract:** A full extract of all historical data (supplier records, equipment certifications, associated documents) will be securely obtained from the legacy SIRIM system. The format is assumed to be a database dump or structured data files (e.g., CSV, XML).
2. **Transform:** The ESB will execute transformation logic to map the legacy data schema to the new NCEF PostgreSQL schema. This includes data cleansing (e.g., standardizing formats), validation, and resolving inconsistencies.
3. **Load:** The transformed data will be loaded into the UAT/Staging environment for initial validation and testing.

Phase 2: Delta Load (After Pilot Launch)

1. After the NCEF system goes live in a pilot phase, the legacy system may still be operational for a grace period. During this time, new records might be created in the old system.
2. A delta migration process will be executed to extract only the new or updated records from the legacy system since the last migration and load them into the NCEF production database. This ensures both systems remain synchronized during the transition.

Phase 3: Final Reconciliation (After Grace Period End)

1. Once the legacy system is ready to be replaced, a final migration run will be performed to ensure any remaining data is transferred.
2. A comprehensive reconciliation process will follow, involving automated scripts and manual checks to verify data completeness and integrity against control totals and checksums.

5.18.3. Interfaces

Integration Hub (IBM webMethods): This is the core component for the migration, containing the ETL logic.

Legacy System: The ESB will connect to the legacy system's database or file share to extract the data for migration.

NCEF Database (PostgreSQL): The ESB will load the transformed data directly into the NCEF database.

Object Storage (MinIO): All associated documents from the legacy system will be migrated to the MinIO object store, and the paths will be updated in the DOCUMENTS table.

5.18.4. Data Requirements

The migration process will populate all relevant tables in the NCEF database, including USERS, COMPANIES, EQUIPMENT, CERTIFICATES, as well as other required columns which will be defined further on another session with SIRIM.

6. HUMAN INTERFACE DESIGN

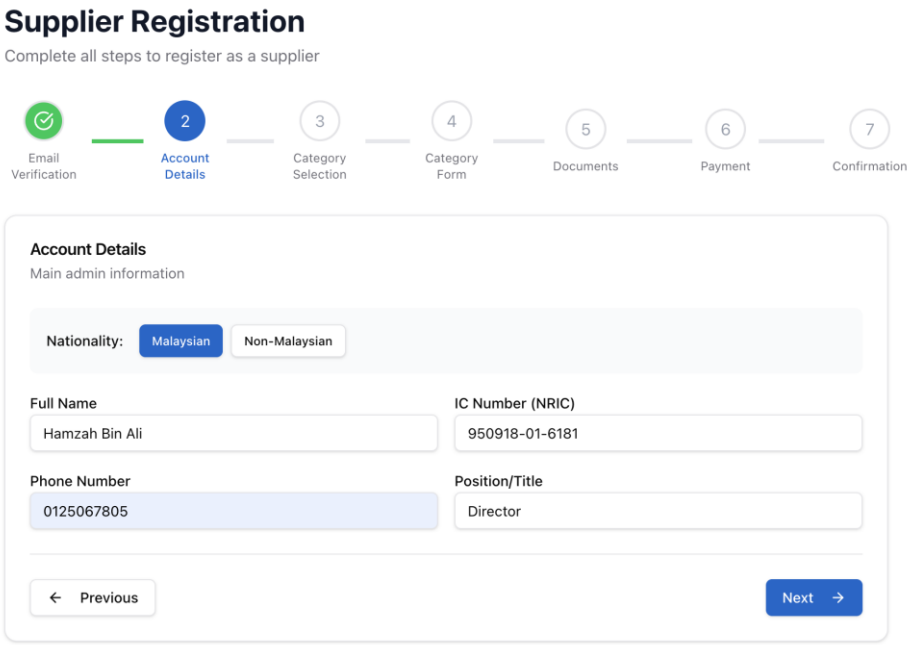
This section outlines the high-level principles and design considerations for the user interfaces of the NCEF system. The goal is to create a user-friendly, accessible, and consistent experience across all portals and applications.

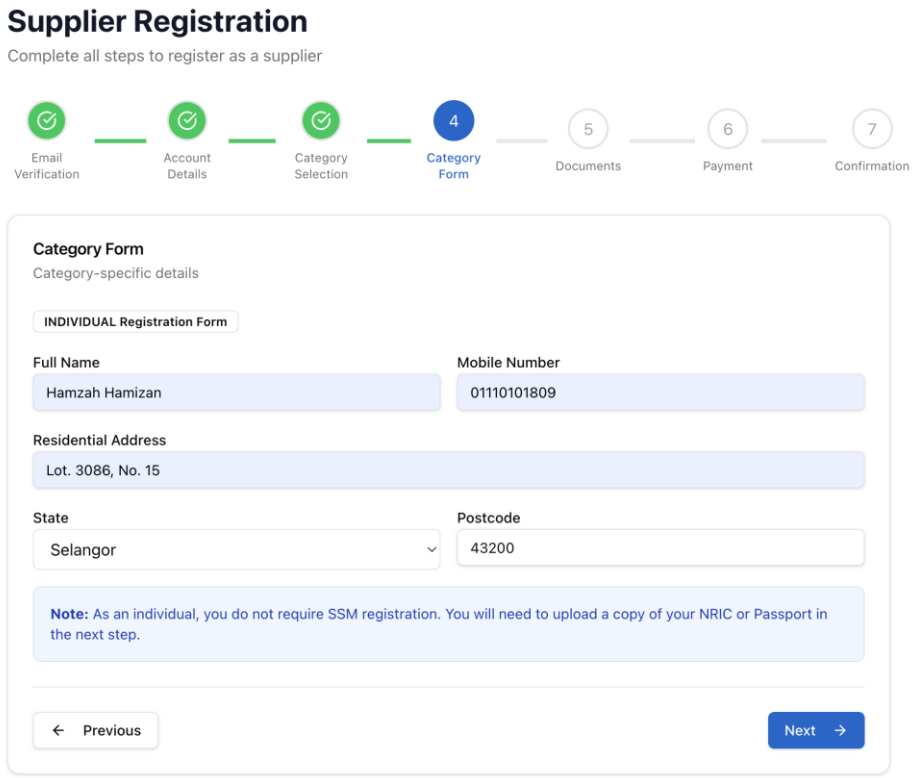
6.1. UI/UX Principles

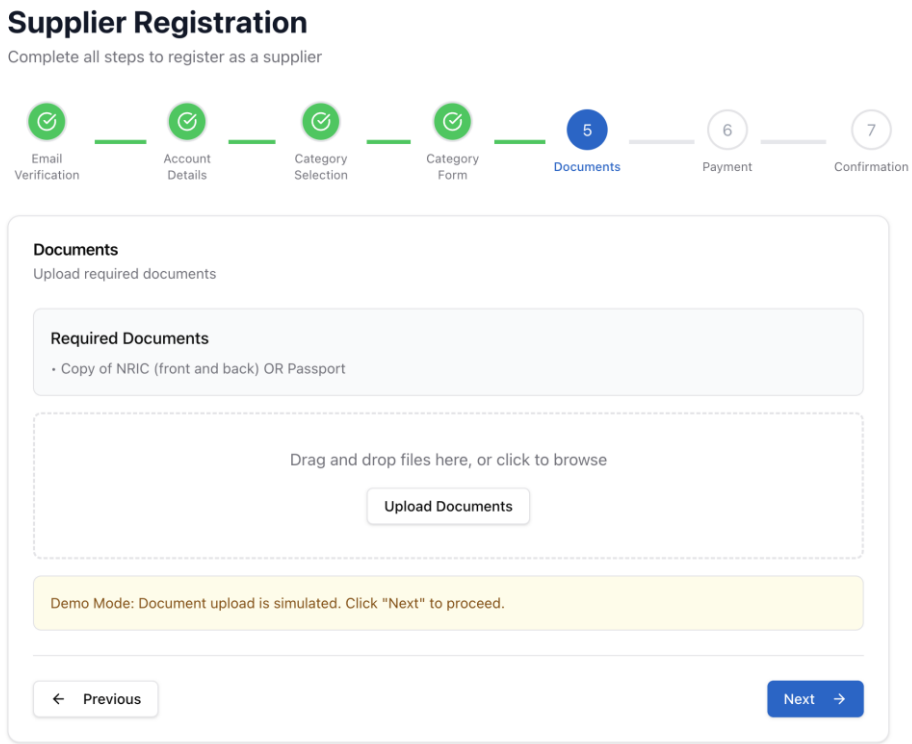
- **Clarity and Simplicity:** The interface will be clean, uncluttered, and intuitive. Complex processes will be broken down into logical, step-by-step wizards to guide the user.
- **Consistency:** A consistent design language (colors, fonts, controls) will be used across the Internal, External, and Public portals to ensure a unified user experience.
- **Responsiveness:** All web interfaces will be fully responsive, providing an optimal viewing and interaction experience on a wide range of devices, from desktop monitors to tablets and mobile phones.
- **Accessibility:** The UI will be designed to comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standards, ensuring it is usable by people with disabilities.
- **Bilingual Support:** All user-facing portals (Public, External, Mobile Application) will be fully bilingual, supporting both English and Malay.

6.2. Key Screen Flows

1. **External User Onboarding:** A multi-step wizard will guide new users through the two-layer registration process, from initial account creation and email verification to detailed profile submission.


Supplier Registration
Complete all steps to register as a supplier
Email Verification
2
Account Details
3
Category Selection
4
Category Form
5
Documents
6
Payment
7
Confirmation
Account Details
Main admin information
Nationality:
Malaysian
Non-Malaysian
Full Name
Hamzah Bin Ali
IC Number (NRIC)
950918-01-6181
Phone Number
0125067805
Position/Title
Director
Previous
Next


Supplier Registration
Complete all steps to register as a supplier
Email Verification
Account Details
Category Selection
4
Category Form
5
Documents
6
Payment
7
Confirmation
Category Form
Category-specific details
INDIVIDUAL Registration Form
Full Name
Hamzah Hamizan
Mobile Number
01110101809
Residential Address
Lot. 3086, No. 15
State
Selangor
Postcode
43200
Note: As an individual, you do not require SSM registration. You will need to upload a copy of your NRIC or Passport in the next step.
Previous
Next



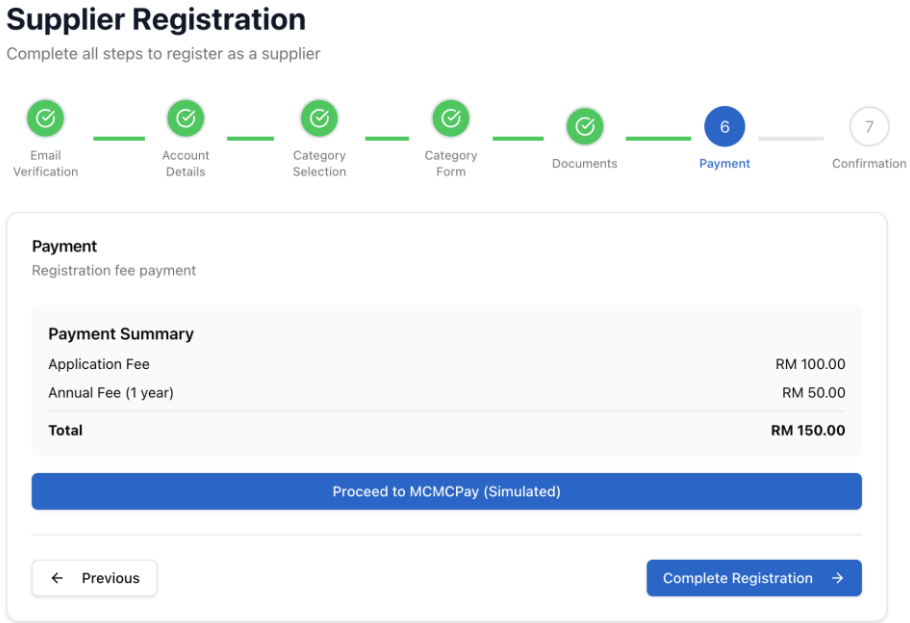


Figure: Example of Supplier Registration Screens
(Terminologies to be updated during UI/UX phase)

2. **SDoC Application Submission:** A tabbed form interface will be used for the SDoC application, separating different sections like Supplier Declaration, Technical Specifications, and Labelling Information.

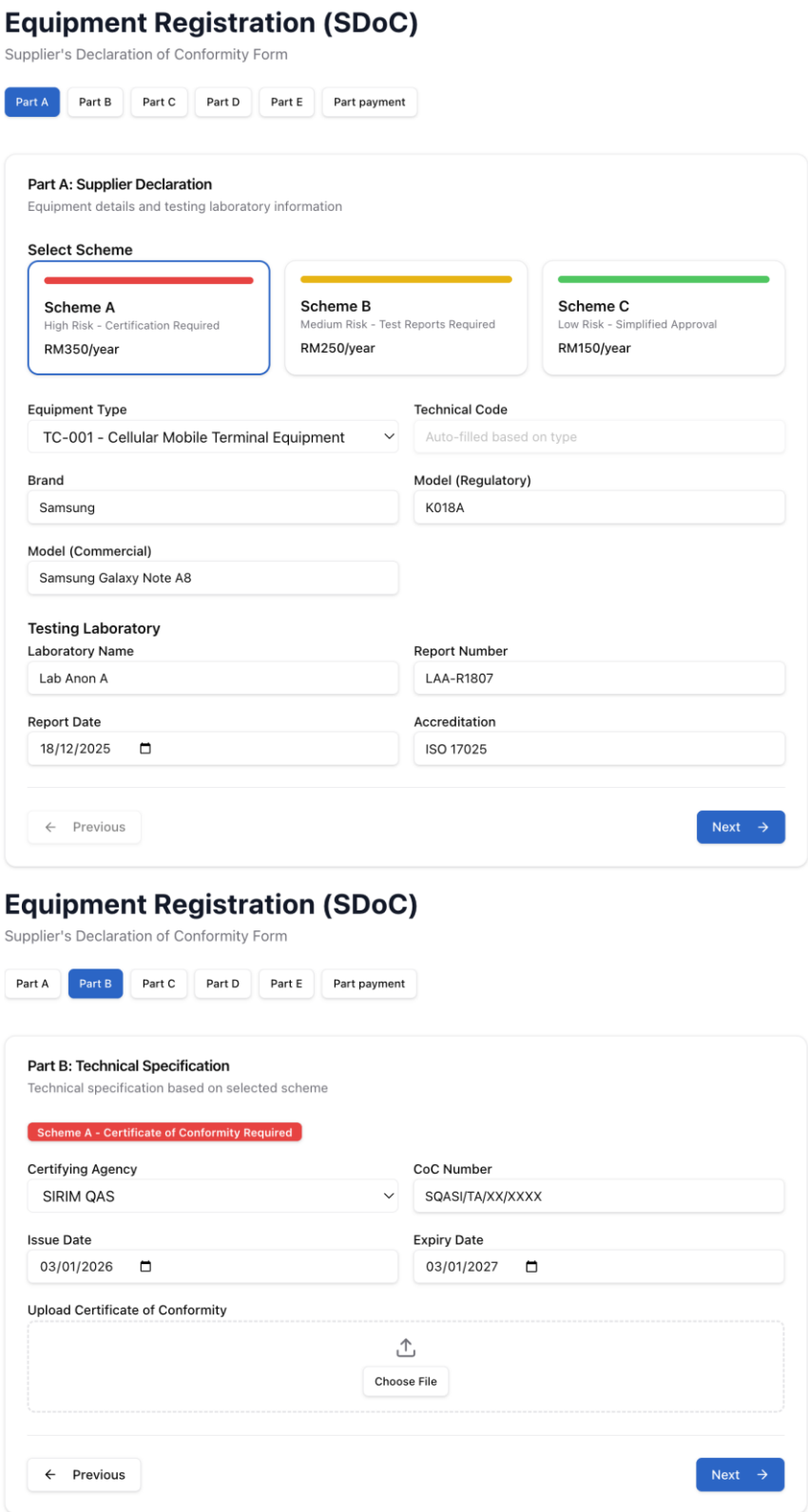


Figure: Example of SDoC Submission Screens
(Terminologies to be updated during UI/UX phase)

3. **Applicant Dashboard:** The main dashboard will provide a summary of key information at a glance, with clear calls-to-action for pending tasks and status updates presented in a sortable, filterable table.

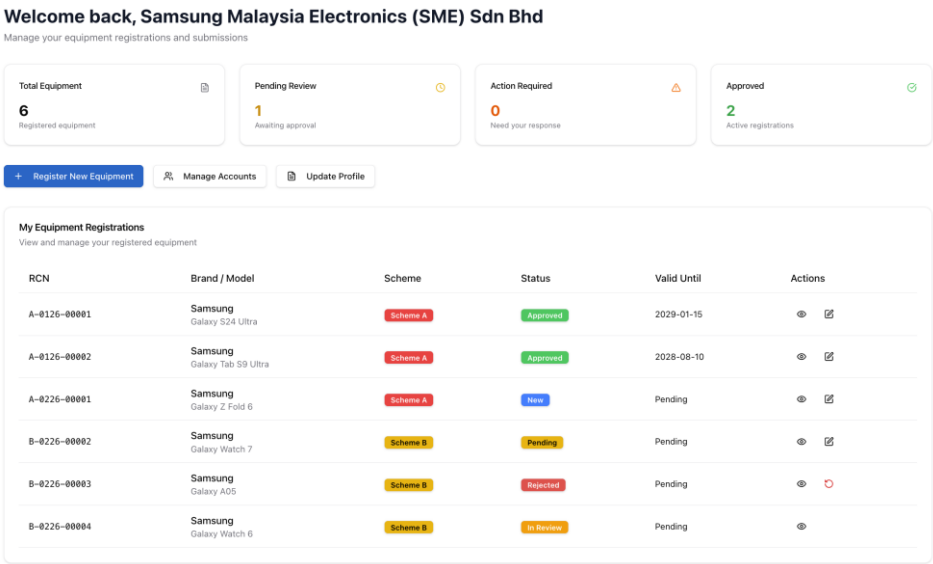


Figure: Example of Application Dashboard Screen
(Terminologies to be updated during UI/UX phase)

4. **Officer Dashboard:** The internal dashboard will be role-based, presenting MCMC officers with a prioritized work queue, SLA timers, and key operational metrics.

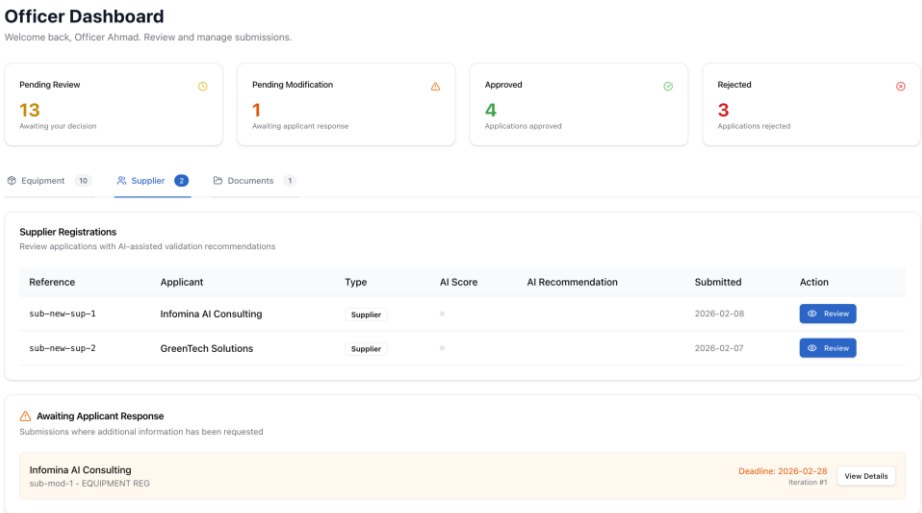


Figure: Example of Officer Dashboard Screen
(Terminologies to be updated during UI/UX phase)

6.3. Wireframes and Mockups

Detailed wireframes and high-fidelity mockups for all key screens will be created in a separate UI/UX design phase and will be appended in a dedicated design repository. The design will adhere to the branding and style guidelines of MCMC.

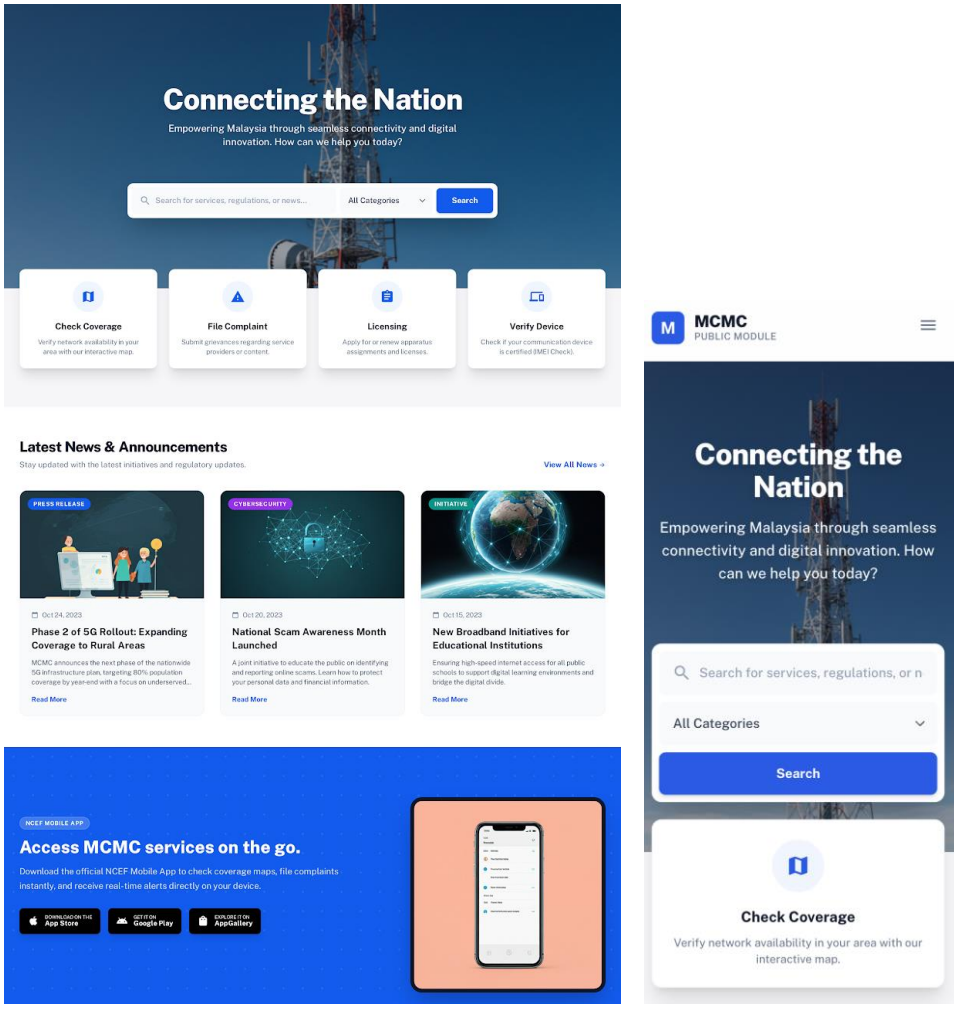


Figure: Example of Public Website and Web Application UI

(Terminologies to be updated during UI/UX phase)

7. NON-FUNCTIONAL DESIGN

This section details the design considerations for the non-functional requirements (NFRs) of the NCEF system, ensuring it meets the required standards for performance, security, availability, and compliance.

7.1. Performance

Requirement: The system must support 500 concurrent users, with page loads under 3 seconds and transaction processing under 5 seconds.

Design:

- **Load Balancing:** Nginx will be used as a load balancer in an active-active configuration to distribute traffic across multiple application server instances.
- **Scalable Architecture:** The microservices architecture, orchestrated by Rancher (Kubernetes), allows for horizontal scaling. Individual services can be scaled independently based on load.
- **Database Performance:** PostgreSQL will be optimized with appropriate indexing. For read-heavy operations like reporting and public search, read replicas or materialized views will be implemented to avoid impacting write performance.
- **Caching:** A caching layer (e.g., Redis, managed within the application layer) will be used to cache frequently accessed data, reducing database load and improving response times.

7.2. Security

Requirement: The system must be protected against common vulnerabilities (OWASP Top 10), enforce strong authentication, and encrypt all sensitive data.

Design:

- **Authentication & Authorization:** Keycloak will manage all identity and access control. It will enforce strong password policies and 2FA for external users, and integrate with Azure AD for secure SSO for internal users. All API endpoints will be secured using OAuth 2.0/JWT access tokens validated by the API Gateway (APISIX).
- **Web Application Firewall (WAF):** Nginx will be configured with a WAF (e.g., ModSecurity) to protect against common web attacks like SQL injection, Cross-Site Scripting (XSS), and CSRF.
- **Data Encryption:**

  - **In Transit:** All communication between the client and the server, and between internal services, will be encrypted using TLS 1.2/1.3.
  - **At Rest:** All sensitive data in the PostgreSQL database and all documents in MinIO will be encrypted at the storage level using industry-standard algorithms like AES-256.
- **Vulnerability Scanning:** Regular static (SAST) and dynamic (DAST) application security testing will be integrated into the CI/CD pipeline to identify and remediate vulnerabilities early.

7.3. Compliance (PDPA)

Requirement: The system must fully comply with the Malaysian Personal Data Protection Act (PDPA) 2010.

Design:

- **Consent Management:** The system will obtain explicit consent from users during registration, with clear links to the privacy policy.
- **Data Minimization:** The system will only collect personal data that is strictly necessary for the registration and regulatory process.
- **Access Control:** Strict RBAC, managed by Keycloak, ensures that only authorized MCMC officers can access personal data, and only for its intended purpose.
- **Audit Trails:** A comprehensive and immutable audit trail will log all access to and modifications of personal data.

7.4. Availability & Reliability

Requirement: The system shall have a target uptime of 99.5%.

Design:

- **High Availability:** The production environment is designed for high availability. Critical components like the Load Balancer, Reverse Proxy, Application Servers, and API Gateway are deployed in an active-active configuration across multiple nodes.
- **Database Redundancy:** The PostgreSQL database will be deployed in an active-passive configuration with automatic failover, ensuring data availability in case the primary node fails.

- **Container Orchestration:** Rancher (Kubernetes) will automatically manage the health of containerized services. If a container or node fails, Kubernetes will automatically restart it or reschedule it on a healthy node.

7.5. Backup & Recovery

Requirement: Daily backups with a 30-day retention policy and disaster recovery environment for continuity.

Design:

Backup Strategy:

- **PostgreSQL:** Automated daily backups will be performed using tools like pg_dump. Backups will be stored securely in a separate location.
- **MinIO:** The object storage will be configured for versioning and replication to a secondary site.
- **Disaster Recovery (DR) Environment:** A DR environment will be deployed.

8. REQUIREMENTS TRACEABILITY MATRIX

This section provides a traceability matrix that maps the requirements specified in the User Requirement Specification (URS) v1.1 to the corresponding sections in this Software Design Document (SDD), ensuring that all requirements have been addressed in the design.

URS Section	Requirement Description	SDD Section(s)
4.1	**User Management Module**	**5.1**
4.1.1	User Classification & Roles	5.1.2
4.1.2	Supplier Registration (External Users)	5.1.2
4.1.3	Principal & Consultant Registration	5.1.2
4.1.4	User Authentication & Login (2FA/SSO)	5.1.2, 7.2
4.1.5	Single Sign-On (SSO) for Internal Users	5.1.2, 3.4
4.1.6	User Profile Management	5.1.2
4.1.7	User Access Matrix	9.1 (Appendix A)
4.1.8	Account Activation & Deactivation	5.1.2
4.2	**Equipment Registration Module (SDoC)**	**5.2**
4.2.1	Scheme A: SDoC with Certification (High Risk)	5.2.2
4.2.2	Scheme B: SDoC with Verification (Medium Risk)	5.2.2
4.2.3	Scheme C: SDoC (Low Risk) - AI Auto-Acceptance	5.2.2, 5.11.2
4.2.4	Equipment Information Requirements	4.2, 5.2.2
4.2.5	Document Upload & Validation	5.2.2, 5.11.2
4.2.8	Declaration & Digital Signature	5.2.2
4.3	**Special Approval Module**	**5.3**
4.3.1	Special Approval - Low/Medium Risk Equipment	5.3.2
4.3.2	Special Approval - High Risk Equipment	5.3.2
4.3.3	Special Approval - Prohibited Equipment	5.3.2
4.3.4	Purpose of Usage Categories	5.3.2
4.3.5	Prohibited Equipment Manual Approval Process	5.3.2
4.4	**Renewal Module**	**5.4**
4.4.1	Supplier/Principal/Consultant Renewal	5.4.2

4.4.2	Equipment Registration Renewal (Scheme A, B, C)	5.4.2
4.4.3	Renewal Period & Validity	5.4.2
4.5	**IMEI/Serial Number Registration Module**	**5.5**
4.5.1	IMEI Registration & Duplication Validation	5.5.2
4.5.2	Serial Number Registration & Duplication Validation	5.5.2
4.5.3	Bulk Registration Capability	5.5.2
4.6	**Modification of Registration Module**	**5.6**
4.6.1	Modification Request Process & Categorization	5.6.2
4.6.2	Acceptance Workflow	5.6.2
4.6.3	Version Control	5.6.2
4.7	**Importation Module**	**5.7**
4.7.2	Integration with RMCD MyOGA System	5.7.2, 3.4, 5.14.2
4.7.3	Trader Information	5.7.2
4.8	**Public Module**	**5.8**
4.8.1	Public Landing Page	5.8.2
4.8.2	Equipment Certification Search & Verification	5.8.2
4.8.3	Guidelines, Standards & Reference Documents	5.8.2
4.8.5	Chatbot	5.8.2, 5.11.2
4.9	**Mobile Application**	**5.9**
4.9.1	Native Mobile Applications (Flutter Approach)	5.9.1
4.9.3	Key Features & Functionality (Search, Push Notifications)	5.9.2
4.10	**Payment Integration Module**	**5.10**
4.10.1	MCMC Pay Integration	5.10.2, 3.4, 5.14.2
4.10.2	Fee Structure & Notice for Payment Generation	5.10.2, 9.2 (Appendix B)
4.10.3	Payment Confirmation & Receipt	5.10.2
4.10.4	Integration with MCMC SIFS	5.10.2, 3.4, 5.14.2
4.11	**AI-Enabled Features**	**5.11**

4.11.1	AI Document Validation (ICR)	5.11.2
4.11.2	Risk Scoring & Assessment	5.11.2
4.11.3	AI-Assisted Decision Support	5.11.2
4.11.4	Auto-Acceptance Thresholds (Scheme C)	5.11.2
4.11.5	Fraud Detection	5.11.2
4.11.6	AI Chatbot/Assistant	5.11.2
4.12	**Dashboard & Reporting Module**	**5.12**
4.12.1	Internal Officer Dashboard	5.12.2
4.12.2	Applicant Dashboard	5.12.2
4.12.3	Analytics & KPI Tracking	5.12.2
4.12.5	Compliance & Operations Reporting	5.12.2
4.13	**Notification Module**	**5.13**
4.13.1	Email Notifications (Internal & External)	5.13.2
4.13.3	System Notifications	5.13.2
4.13.4	Push Notifications	5.9.2, 5.13.2
4.13.5	Notification Templates	5.13.2
4.14	**Integration Requirements**	**3.4, 5.14**
4.14.1	SIRIM eComM System Integration	5.14.2
4.14.6	Data Migration from Legacy SIRIM System	5.16
4.15	**Configuration & Settings**	**5.15**
4.15.1	System Parameters	5.15.2
4.15.3	Officer Grouping & Assignment / Calendar Blocking	5.15.2
4.15.4	Fee Structure Configuration	5.15.2, 9.2 (Appendix B)
4.16	**General System Requirements**	**4.0, 7.0**
4.16.1	Data Validation Rules	4.1, 4.2
4.16.2	Audit Trail & Logging	7.2, 7.3
4.16.3	Document Management	4.1, 4.2 (MinIO)
5.0	**Non-Functional Requirements**	**7.0**
5.1	Performance Requirements	7.1

5.2	Security Requirements	7.2
5.3	Compliance Requirements (PDPA)	7.3
5.4	Availability & Reliability	7.4
5.5	Scalability	7.1
5.8	Backup & Recovery	7.5

9. APPENDICES

Appendix A: User Access Matrix

This appendix provides a high-level overview of the access rights for the primary user roles within the NCEF system. A more detailed, field-level permission matrix will be maintained in Keycloak. As for officer, System Admin will be able to appoint OIC lead which will enable additional ability to manually override applicant assignment.

Module	System Admin	OIC / Verifier / Recommender	Acceptance Status	Applicant (External)	Public (Unauthenticated
User Management	C R U D	R	R	R (Own)	-
Equipment Registration (SDoC)	R	C R U	R U	C R U (Own)	-
Special Approval	R	C R U	R U	C R U (Own)	-
Renewal	R	R U	R U	C R U (Own)	-
Modification	R	R U	R U	C R U (Own)	-
IMEI/Serial Number Registration	R	R	R	C R U (Own)	-
Importation	R	R	R	C R (Own)	-
Public Module	C R U D	C R U D	R	R	R
Dashboard & Reporting	R (All)	R (Assigned)	R (All)	R (Own)	-
Configuration & Settings	C R U D	-	-	-	-

Legend: C = Create, R = Read, U = Update, D = Delete

Appendix B: Fee Structure

This appendix summarizes the fee structure for the NCEF system, as defined in the URS [1]. All fees are configurable by the System Administrator.

Table 1: User Registration Fees

Category	Application Fee	Annual Fee	Max Period
All (Company, Individual, etc.)	RM 100.00	RM 50.00 / year	5 years

Table 2: Equipment Registration Fees (SDoC)

Scheme	Fee per Year
Scheme A (High Risk)	RM 350.00
Scheme B (Medium Risk)	RM 250.00
Scheme C (Low Risk)	RM 150.00

Table 3: Special Approval Fees

Category	Fee per Unit
Low/Medium Risk (Registered Model)	RM 20.00
Low/Medium Risk (Non-Registered Model)	RM 40.00
High Risk Equipment	RM 300.00
Prohibited Equipment	RM 300.00

Table 4: IMEI and Serial Number Registration Fees

Type	Fee per Unit
IMEI Number	RM 0.50
Serial Number	RM 0.15

Table 5: Renewal Fees

Type	Fee
User Registration Renewal	Same as initial registration (Annual Fee only)

Equipment Registration Renewal	Same as initial SDoC registration fees

Appendix C: Proposed AI Scoring Mechanism:

The following scoring criteria are used to determine the confidence score:

Criteria	Weightage	Description
SSM/ROC/ROB Validation	20%	Successful real-time validation with the SSM database confirms a legitimate business entity.
Completeness of Technical Specifications	15%	All required technical specifications are provided.
Test Report & Lab Verification	15%	Test report is from a recognized, accredited lab with a valid report number.
Address & Director Match	10%	Information matches official SSM records, reducing fraud risk.
Brand & Model Consistency	10%	Brand and model are consistently named across all documents and not on any blacklist.
Standards Compliance	10%	Equipment is tested against the latest MCMC-required technical standards.
Supplier's Past Performance	10%	Supplier has a good track record of compliant applications.
Completeness & Consistency	10%	All fields are correctly filled with no contradictions.
Total Score	**100%**	

Auto-Acceptance Threshold, which only applies for Scheme C and Renewal:

Score Range	Application Status	Action Detail
90% and above	**Automatic Acceptance** *(Strictly for Scheme C and Renewal)*	**Scheme C/Renewal:** The application is accepted instantly without manual intervention. **Other Schemes/Special Approval:** The application score is displayed to the officer, with justifying reasons based on the criteria listed in the previous table. *As for the remainder 10% of the missing score, for scheme C/Renewal it will be taken for cumulative of each criteria, therefore when one criteria worth 10% is missing entirely (*eg. Address & Director match*), the system shall still request for review.
70% – 89%	**Priority Review**	Flagged for an MCMC officer; lower-scoring sections are highlighted for targeted assessment.
Below 70%	**Standard Manual Review**	Sent to the general queue for a comprehensive, manual evaluation by an officer. Lower-scoring sections as well as invalid/missing/tampered documents are highlighted for targeted assessment.